

Martin-Luther-Universität Halle-Wittenberg
Faculty of Natural Science II
Institute of Physics

Physics Modelling Assistant

Using Agentic Artificial Intelligence in Physics Modelling

Bachelor's Thesis

for the Award of the Academic Degree
Bachelor of Science
in Physik und Digitale Technologien

submitted by: Janis Felix Jacobi

Supervisor 1: Prof. Dr. Niels Schröter

Supervisor 2: Prof. Dr. Samir Lounis

submission: Halle (Saale), 24th August 2026

Contents

Table of Figures	4
Table of Tables	4
Glossary	4
1 Introduction	5
2 Theoretical Foundations	6
2.1 Definitions	6
2.1.1 Artificial Intelligence	6
2.1.2 Large Language Model	6
2.1.3 Agentic AI	6
2.2 Decoding the Output of LLMs	7
2.2.1 LLM as a Function which Assigns Probabilities	7
2.2.2 Greedy Decoding	8
2.2.3 Softmax Decoding	8
2.2.4 Top-k Decoding	8
2.2.5 Top-p Decoding	9
2.3 Other Parameters which Can Influence an AI System's Performance	9
2.3.1 Selection of the LLM	9
2.3.2 Tool Use of LLMs	10
2.3.3 Reasoning	10
2.4 Orchestrating AIs to Behave as an Agentic AI	11
2.4.1 Orchestration Architectures	11
2.4.2 Communication Inside Agentic AI Systems	12
2.4.3 Memory	12
2.5 Evaluation of AI Systems	12
2.5.1 CritPt	13
3 Design of the Physics Modelling Assistant	13
3.1 Orchestration Framework: CrewAI	13
3.2 LLMs Used	15
3.3 Tools Used	15
3.3.1 arxiv_paper_tool with Exponential Backoff	16
3.3.2 pdf_reader	16
3.3.3 dimensional_analysis	16
3.3.4 Code Execution	17
3.4 Experimental Setup for the Pipeline	17
3.4.1 Input	17
3.4.2 General Parameter Choices	17
3.4.3 Obtaining Information	18
3.4.4 Building the Model	19
3.4.5 Implementing the Model	19
3.4.6 Giving a Final Answer	20
3.4.7 Output	20
3.4.8 Connecting to CritPt's Benchmark	21

3.5	Investigating the Physics Modelling Assistant's Performance	21
3.5.1	An Example Run	21
3.5.2	Pipeline's Performance Compared to Individual Models	22
3.5.3	The Relevance of the Available Sources	22
3.5.4	The Effect of Iterations of the Agents	22
4	Results and Discussion	23
4.1	Results of the Example Run from the Reference Setup	23
4.1.1	Discussion of the Results from the Example Run	30
4.2	Evaluation of the Physics Modelling Assistant	30
4.2.1	Issues Encountered during the Generation	30
4.2.2	Benchmarking Results for the Default Setup	30
4.2.3	Benchmarking Results when Using DeepSeek V4 Flash	30
4.2.4	Benchmarking Results without Accessing External Sources	31
4.3	General Discussion of Benchmarking Results	31
5	Conclusion and Future Work	32
6	Appendix	39
7	Declaration of authorship	40
8	Acknowledgement	41

Figure Index

CrewAI execution modes	14
Overview of the Physics Modelling Assistants Tasks	20
Simulation Output	29

Table Index

Benchmarking SAIA models	15
------------------------------------	----

Glossary

Agent a singular AI system which is commonly part of an Agentic AI system 3–5, 11–20, 22, 25, 27, 32

Agentic AI an AI system with enhanced capabilities (subsubsection 2.1.3) 2, 4–6, 10–12, 32

AI Artificial Intelligence 2, 4–7, 9–13, 32

API key application programming interface key; a unique and secret identifier granting access to a service 4, 14, 15, 21

context-window a size describing how many previous output tokens a LLM can take into consideration 4, 6, 11, 16–18, 30, 32

LLM Large Language Model 2, 4, 6–20, 22, 30, 32

logit a LLM’s output for every token indicating how well it is suited to be the next token. Larger values indicate a better fit. [47, p. 4] 4, 7, 8

token the input for a LLM’s transformation function 4, 6–11, 14, 16–18

1 Introduction

Artificial Intelligence can be of great assistance in scientific research [72]. As shown by Chugunova et al. [15], researchers in natural science are among the groups most familiar with AI compared to other scientists. There is a wide range of applications, stretching from finding sources in the literature to aiding in the presenting research findings. AI can be useful for forming hypothesis and evaluating them by designing and controlling practical experiments and aiding during simulations, especially with the program implementation. [72]

In the past two years automated Agentic AI research pipelines were developed with the potential to improve the work with AI in science. The first fully autonomous approach was published by Lu et al. in 2024 [54], introducing a framework called “The AI Scientist” [54]. It identifies interesting research questions, plans and executes according experiments, interprets the results, and writes a paper stating its findings via a multistep process. A similar system is the “InternAgent” [69], which provides an interface for introducing human feedback into the system. This is realised by the “Orchestration Agent” [69] managing how and when human input is introduced into the system. The system itself consists of the four distinct tasks of “Survey”, “Code Review”, “Assessment”, and “Idea Innovation” [69]. The “Orchestration Agent” [69] manages the interactions between the tasks and how the findings are presented to the human. This design shows some resemblance to human research groups with individuals specialized in specific aspects of the work while a supervisor coordinates the process. In tests ran by the “InternAgent” developers [69], their system was able to outperform the “AI-Scientist-V2” [77], a successor of “The AI Scientist” [54].

The work on the “InternAgent” [69] also introduced tracking the cost of the AI usage during a run of the system. In this metric the “InternAgent” [69] outperformed the “AI-Scientist-V2” [77] again, indicating that higher costs do not necessarily yield better results.

A less automated approach was made by Shao et al. with the introduction of “SciSciGPT” [67]. This system is developed for Science of Science research, which is why “SciSciGPT” is equipped with data management component amongst a literature review and an analytics component. Similar to the “InterAgent”, some form of orchestrator is used. It receives the outputs of the subtasks along with an evaluation from a task designated to critique the outputs of other tasks. In contrast to the “AIScientist” [54] and “InternAgent” [69], “SciSciGPT” [67] does not generate its own research ideas. Instead it responds to an input prompt, allowing for seamless integration into the work of human scientists.

As of 1st June 2026, the latest advancement in the field was made by Louapre, Niklaus, and Tunstall presenting the “physics-intern” [53], “a multi-agent scaffolding system that takes a physics problem and works through it autonomously” [53, section 3]. Their approach is similar to previous systems, using an orchestrator to manage a research and computation component. The results are reviewed and checked against the literature, before the critique is passed to the “Strategy Planner” [53], which has access to a literature “Background Survey” [53] and can adjust the orchestration strategy. This process is repeated until the review deems the result correct or the maximum number of iterations is reached. The “physics-intern” [53] was able to exceed the performance of the models used for its Agents by over 75%, highlighting the potential of orchestration frameworks [53].

This Thesis aims to develop a system with functionality similar to the “physics-intern” [53] (at the time the work on this Thesis began, the “physics-intern” [53] was not yet published). The Physics Modelling Assistant receives a physics problem or modelling request and crafts an according model along with its implementation in order to answer the request. The performance of the system is measured to provide inside into its usefulness. In contrast to the “physics-intern” [53], the Physics Modelling Assistant uses comparably weaker models, aiming to explore if improvements similar to the “physics-intern”’s [53] can be achieved. The influence of

some of the system's configurations is explored by measuring their effect on the performance, the focus is on the effect of external information. With this work I want to contribute to a better understanding of Agentic AI's parameters and a more efficient use of AI systems in scientific research.

2 Theoretical Foundations

Before going in to the details of AI-based systems, some key terms are be defined.

2.1 Definitions

2.1.1 Artificial Intelligence

Artificial Intelligence (AI) describes the functionality of a system. There is no universally accepted definition for AI, so for the purpose of this Thesis, the definition from Russell and Norvig [66, p. 22-26] is used. They describe AI as being a rational actor (rational agent), which acts in a way that achieves the best result based on the expectation of what the result should be ([66, p. 26], translated). This definition allows for a variety of systems to be considered an AI and accounts for the uncertainty regarding the goal the system should work towards. Therefore, classical PID-controllers and software for processing and displaying data, as crafted by Jacobi [46], could be considered AI systems.

2.1.2 Large Language Model

At its core a Language Model is a "Transformer Language Model" [79] as described by Zhao et al. It operates on small words or parts of words, which are called tokens.

When a LLM receives an input string, it divides it into tokens and puts into the transformation function. Based on these input tokens and the previous output tokens, the model calculates the most likely next output token. The number of tokens a model can see, and therefore consider, is limited, the limit is called the context-window of the model [50]. The final result of the LLM is its calculation for most likely sequence of output tokens given the provided input, which means, it provides the best result for the given goal, thereby fulfilling the definition of AI (see subsection 2.1.1). The process of developing capable LLMs will not be discussed in this Bachelor's Thesis.

The distinction between Language Models and Large Language Models is in the number of internal parameters a model has. As stated in *A Survey of Large Language Models* [79], there is no hard cut-off for a Language Models to be considered Large. Commonly, models which are considered Large Language Models have $\geq 10 \cdot 10^9$ parameters.

2.1.3 Agentic AI

As described by Bandi et al., Agentic AIs are "autonomous, goal-driven systems that can operate on their own for long periods, requiring minimal human supervision" [11, section:3. Results]. In order to achieve

this behaviour, these systems combine abilities which are beyond LLMs. Bandi et al. highlight the “strategic planning” [11, section:3. Results] capabilities, the preservation of the tasks context over time [11], the usage of “external tools to extend [the systems] capabilities” [11] and the collaboration of individual Agents (this means AI-systems) with other Agents. [11]

2.2 Decoding the Output of LLMs

In subsection 2.1.2, LLM is defined based on the functionality it provides. In order to understand how the behaviour of LLMs can be modified, a more formal definition is given, based on the work of Ji et al. in *Decoding as Optimisation on the Probability Simplex: From Top-K to Top-P (Nucleus) to Best-of-K Samplers* [47].

2.2.1 LLM as a Function which Assigns Probabilities

A LLM can be described as a function f with internal parameters θ . Additionally, f takes a set of input token $v_{in} = (i_1, \dots, i_N)$, where $N \in \mathbb{N}$ is the number of input tokens, and a set of previous output tokens $v_{out,prev} = (o_{-1}, \dots, o_{-M})$, where $M \in \mathbb{N}$ is the number of previous output tokens considered. The function $f(\theta, v_{in}, v_{out,prev})$ assigns a probability $s(v)$ to every token $v \in \mathcal{V}$, where $\mathcal{V}$ is the vocabulary of tokens known to the LLM:

$$f(\theta, v_{in}, v_{out,prev}) = \begin{pmatrix} s(v'_1) \\ \dots \\ s(v'_Z) \end{pmatrix} = s(v) \quad \text{where } Z \text{ is the size of the vocabulary } \mathcal{V} \quad (2.2.1)$$

v'_i with $i \in [1, Z] \subset \mathbb{N}$ is an element from the ordered set of $\mathcal{V}'$, which allows to map the rows of the probability matrix $s(v)$ to the tokens of the vocabulary. The probability indicates how well a token is suited to be the next output token [47, p. 4]. The probabilities $s(v'_i)$ are called “logits” [47, p. 4], they are not normalized by default and larger values of a logit signal a better fit for the corresponding token to be the next output token.

The LLM by itself outputs a probability distribution including every token from its vocabulary. To choose a token based on this probability distribution, Ji et al. state that the following optimization problem needs to be solved, they label it the “MASTER PROBLEM” [47, p. 5]:

$$q^* = \arg \max_{q \in \Delta(\mathcal{V})} \left[\underbrace{\langle q, s \rangle}_{\sum_{v \in \mathcal{V}} q(v) \cdot s(v)} - \lambda \Omega(q) \right] \quad (2.2.2)$$

The used quantities are:

- q^* is the probability distribution maximizing Equation 2.2.2
- $\Delta(\mathcal{V})$ is the set of all possible probability distributions
- q is a single probability distribution for a given vocabulary $\mathcal{V}$
- s is the initial vector of logits the LLM produces
- $\Omega(q)$ is a regularisation function, which can push q^* towards a certain distribution
- λ is the strength of the regularisation function, with $\lambda \geq 0$

There are different approaches for finding q^* and eventually getting a single token from it, the once relevant to this Thesis are discussed in the following.

2.2.2 Greedy Decoding

This method uses no special regularisation function $\Omega(q)$, and therefore sets $\lambda = 0$. This simplifies Equation 2.2.2 to:

$$q^* = \arg \max_{q \in \Delta(\mathcal{V})} \langle q, s \rangle \quad (2.2.3)$$

The resulting q^* vector has all the probability mass placed on the token, which was calculated to be most likely, while all other tokens get assigned a probability of 0. Therefore, the vector q^* has value 0 in every row except the one corresponding to the most likely token, which has probability 1. This means the next token is always the most likely token according to the LLM's calculations. If multiple tokens have the highest probability $s(v)$, a single token is selected amongst them. [47, p. 10].

2.2.3 Softmax Decoding

For Softmax Decoding, the regularisation function is chosen as $\Omega(q) = -\sum_{v \in \mathcal{V}} q(v) \log(q(v))$, which is the negative entropy of q . Using this $\Omega(q)$ in solving Equation 2.2.2 leads to the following optimal distribution q^* , where $q_t^*(v)$ is the probability per token: [47, p. 10-12]

$$q_t^*(v) = \frac{\exp(\frac{s(v)}{\lambda})}{\sum_{u \in \mathcal{V}} \exp(\frac{s(u)}{\lambda})} \quad (2.2.4)$$

This distribution is normalised and has the parameter λ , which controls the spread of the probability distribution q^* . For larger values of λ , the probability is spread more evenly across all tokens, which results in previously unlikely tokens being more probable than before. When λ becomes very small, the probability distribution is sharper, with the probability mass being concentrated more around a few tokens which already were likely before. The next output token is selected according to the final probability distribution. For the edge case $\lambda = 0$, the regularisation function is disabled (see Equation 2.2.2), which causes the Softmax Decoding to become Greedy Decoding (see subsection 2.2.2). [47, p. 10-12]

The parameter λ is one way of controlling the general randomness of the output. In the context of LLMs, this parameter is often associated with the creativity of a model and called **Temperature** T . [47, p. 10-12]

2.2.4 Top-k Decoding

Top-k Decoding extends Softmax Decoding (subsection 2.2.3) by selecting the k ($k \in \mathbb{N} \setminus k = 0$) tokens with the highest logit values [47, p. 12]. These tokens form a subset of the vocabulary $\mathcal{V}$ called $\mathcal{V}_k$ on which Softmax Decoding is performed, while the probability for every other token $v \in \mathcal{V} \setminus \mathcal{V}_k$ is set to 0. This gives the following probability distribution [47, p. 12]:

$$q_t^*(v) = \begin{cases} \frac{e^{\frac{s(v)}{\lambda}}}{\sum_{u \in \mathcal{V}_k} e^{\frac{s(u)}{\lambda}}} & \text{for } v \in \mathcal{V}_k \\ 0 & \text{else} \end{cases} \quad (2.2.5)$$

For $k \geq |\mathcal{V}|$, Top-k Decoding is equivalent to the standard Softmax Decoding (see subsection 2.2.3). Setting $k = 1$ results in the same functionality as Greedy Decoding (see subsection 2.2.2). [47, p. 12]

From Top-k Decoding the parameter **top_k** is derived, which can be used to shape the output of LLMs [47, p. 12].

2.2.5 Top-p Decoding

Top-p Decoding is another extension to Softmax Decoding (subsection 2.2.3), also selecting a subset of all tokens called $\mathcal{V}_p$ from the vocabulary $\mathcal{V}$. In contrast to Top-k Decoding (subsection 2.2.4), Top-p Decoding defines a certain amount of probability $p \in (0, 1]$, on which Softmax Decoding is performed. This is achieved by selecting the token with the highest probability $s(v)$ from the set $\mathcal{V} \setminus \mathcal{V}_p$ and adding it to the set $\mathcal{V}_p$, thereby removing it from $\mathcal{V} \setminus \mathcal{V}_p$. This is repeated until the sum of the probabilities of all tokens $v \in \mathcal{V}_p$ exceeds the value of p . The probability of the tokens in $\mathcal{V} \setminus \mathcal{V}_p$ is set to 0, while Softmax Decoding (subsection 2.2.3) is performed on the tokens in $\mathcal{V}_p$. The resulting probability distribution is ([47]):

$$q_t^*(v) = \begin{cases} \frac{e^{\frac{s(v)}{\lambda}}}{\sum_{u \in \mathcal{V}_p} e^{\frac{s(u)}{\lambda}}} & \text{for } v \in \mathcal{V}_p \\ 0 & \text{else} \end{cases} \quad (2.2.6)$$

For $p = 1$, Top-p Decoding is equivalent to Softmax Decoding, since all tokens are included in $\mathcal{V}_p$. For $p \rightarrow 0$, only the token with the highest probability gets a non 0 probability, which makes this case equivalent to Greedy Decoding (see subsection 2.2.2). [47]

Top-p Decoding allows for dynamic adaptation based on how sharp or spread out the probability distribution returned by the LLM is. For a sharp probability distribution, only the few very likely tokens keep their calculated probabilities and Softmax Decoding (subsection 2.2.3) is performed on them. Whereas, for a flatter distribution, a higher number of tokens are included into the Softmax Decoding (subsection 2.2.3) and therefore are possible output tokens. [47]

The derived parameter is called **top_p** and is commonly used in the field of LLMs to modify a system's output [47, p. 13].

2.3 Other Parameters which Can Influence an AI System's Performance

As shown by numerous benchmarks ([41], [42]), different LLMs achieve different performances. Some of the important factors for these differences are discussed in the following.

2.3.1 Selection of the LLM

One of the most noticeable differences in LLMs is the number of internal parameters they use. The general assumption is that more parameters lead to better performance. Another important feature is the amount of data used to train a LLM. As shown by Kaplan et al. [48, p. 5], combining the number of internal parameters

N a model uses, and the number of tokens D used in training a model can predict the “cross-entropy loss” [48, p. 6] L of a LLM:

$$L(N, D) = \left[\left(\frac{N_C}{N} \right)^{\frac{\alpha_N}{\alpha_D}} + \frac{D_C}{D} \right]^{\alpha_D} \quad (2.3.1)$$

The used quantities are:

- $L(N, D)$ is the “cross-entropy loss” [48, p. 6]. Lower Values indicate a better model.
- N_C is the critical number of internal parameters, until which Equation 2.3.1 is applicable. $N_C \approx 8.8 \cdot 10^{13}$ [48]
- N is the number of internal parameters the model uses.
- D_C is the critical number of tokens, until which Equation 2.3.1 is applicable. $D_C \approx 5.4 \cdot 10^{13}$ [48]
- α_N is the exponent for predicting $L(N)$. $\alpha_N \approx 0.076$ [48]
- α_D is the exponent for predicting $L(D)$. $\alpha_D \approx 0.095$ [48]

There are also other design and training related influences on a models performance, Kaplan et al. discuss the relevance of the number of training steps or the batch size (tokens per step) [48]. Liu et al. [52] show the importance of how the training data are composed, highlighting the increases in predicting a models performance when considering this factor.

2.3.2 Tool Use of LLMs

Since all LLMs have a limited amount of training data, for every model there are domains it lacks basic knowledge in, hindering it to perform tasks in this domain effectively. In order to mitigate this, LLMs are fed extra information to help them answer the question at hand. Realizing this via human input is inefficient and impractical, especially when the human itself is not very knowledgeable in the domain at hand. For these reasons, LLMs are getting equipped with tools, allowing them to access information, which is not included in their training data. As shown by Komeili, Shuster, and Weston [49] and Thoppilan et al. [71], equipping a LLM with some sort of web search tool allows to reduce the false outputs a system produces. Note that tool use is one of the features making an AI an Agentic AI, as discussed in subsubsection 2.1.3.

Another common class of tools are file readers, they can be able to parse various formats like PDF, JSON, or plain text.

Code execution tools allow a LLM to run generated code autonomously. This introduces the problem of untrusted code execution, since the model could generate malicious code, accessing private information or modifying important data on the system it executes the code on. Executing untrusted code should therefore be done in a container separate from important data and systems. As stated by Marchand et al. [55], these containers, called Sandboxes, are not always fully secure; strong LLMs are able to escape them sometimes.

2.3.3 Reasoning

In the context of LLMs, reasoning is creating a chain-of-thought, which means partitioning one big task into several smaller subtasks. A chain-of-thought improves the performance of LLMs on tasks which require a multi-step process to obtain the correct solution, as shown by Wei et al. [76].

2.4 Orchestrating AIs to Behave as an Agentic AI

As mentioned in subsubsection 2.1.3, combining multiple AI systems into one enables the resulting system to perform more complex tasks and achieve better results. The individual AI systems of an Agentic AI system are commonly referred to as Agents. In the following, the main approaches on how Agents can work together are presented, as there where outlined by Kulkarni and Kulkarni [51].

2.4.1 Orchestration Architectures

The Sequential Pipeline [51, p. 3] is a linear chain, where each Agent is executed one after another [51]. Individual Agents or their tools can be rerun, as long as the pipeline is still in the corresponding step. But once an Agent is done, it can not be called again (in the same run). The design process is relatively simple and the resulting system show very deterministic behaviour, which allows errors to be located easily. [51]

The Parallel Fan-Out with Merge [51, p. 4] differs from the Sequential Pipeline (paragraph 2.4.1), by allowing certain Agents to run simultaneously. After the parallel Agents are done with their tasks, their results are merged by the next sequential Agent. Introducing an Agent before the parallelization allows for each parallel Agent to only receive the information relevant to its task. Therefore, each parallel Agent has to deal with less noise and receives less input tokens, which helps to stay within the LLM's context-window. The Agent responsible for merging resolves conflicts like contradicting information of parallel Agents. The parallel design allows for the highest throughput compared to the other architectures [51, p. 7].

The Hierarchical Supervisor-Worker [51, p. 4-5] architecture introduces a supervisor Agent. It dynamically decides which Agents (Subagents) to call and which tasks the Subagents should handle. Furthermore, the supervisor Agent reviews the Subagents' results and reruns the respective tasks, if it has low confidence in the result being correct. This forms a feedback loop, which allows the Agentic AI system to dynamically assign more difficult tasks to more powerful Agents. The hierarchical architecture produces better results than the Sequential Pipeline (paragraph 2.4.1) and the parallel design (paragraph 2.4.1), but is also more costly regarding the execution time. [51, p. 7]

The Reflexive Self-Correcting Loop [51, p. 5] executes the Agents in a predefined order similar to the Sequential Pipeline (paragraph 2.4.1), but adds a verification step. The verification is performed by an Agent, which either deems the result to be correct (enough) and declares it to be the system's output, or sends it back for correction. Before the result is reintroduced into the system, it is critiqued by stating the problems encountered, which allows for targeted corrections on the next cycle. This architecture refines the output over time rather than simply rerunning the entire system. The Reflexive Self-Correction Loop [51, p. 5] achieves the highest accuracy but also comes at the highest cost of all architectures discussed. [51, p. 7]

When interpreting the performances of the architectures discussed, it should be taken into account that Kulkarni and Kulkarni [51, p. 14-15] tested them on their extraction capabilities from financial documents. Therefore, the performances could change in different domains and for different tasks. When designing Agentic AI systems, the combination of multiple architectures is possible.

2.4.2 Communication Inside Agentic AI Systems

Communication protocols are used to define how the different components of an Agentic AI system interact. An orchestration framework may implement some or all of these protocols or similar ones. The LLMs are wrapped into a layered architecture, which handles the interactions between different LLMs and manages how tools are called. [32]

The Model Context Protocol (MCP) [29] (documentation: [70]) handles how the underlying LLM of an Agent uses tools.

The Agent Communication Protocol (ACP) [29] (documentation: [27]) allows different Agents to communicate with each other. It is based on a client-server architecture and uses “RESTful” communication [34].

The Agent-to-Agent Protocol (A2A) [29] (documentation: [68]) handles Agent interactions to enable effective cooperation. This is achieved by Agents communicating their capabilities to each other [34].

The Agent Network Protocol (ANP) [29] (documentation: [14]) manages decentralised communication between Agents. It allows for secure interactions over the internet [34].

2.4.3 Memory

In the context of Agentic AI, memory is an Agent’s awareness of previous actions. Short-term memory is the context of the task the Agent is currently performing. Long-term memory allows an Agent to access information it obtained during previous runs [29]. Other kinds of memory include “semantic memory” [29], “procedural memory” [29], and “episodic memory” [29], but these will not be discussed in this Thesis.

2.5 Evaluation of AI Systems

In order to measure the influence of an AI system’s configurations, a metric for the systems performance is required. One approach is testing the AI system on a set of multiple choice questions, with each answer being clearly right or wrong [40]. The total score over the set of questions indicates the system’s performance. This method is used for the “GPQA Diamond” benchmark [65] and partly for the “Massive Multi-discipline Multimodal Understanding and Reasoning (MMMUR)” [78] benchmark, which are both part of the Artificial Analysis *LLM Leaderboard* [2].

Another option are open answer questions, which require a written response by the tested model. This method is used by “SciBench” [73], where the answers usually are short and need to match the expected answer exactly, since a string comparison of them is performed. The performance over all questions is accumulated and provides a final evaluation score of the tested system.

In order to properly assess an AI system’s capability to handle more complex tasks, human grading can be employed. This is done by Arena Intelligence [1], where users write a prompt and pick the best response from two anonymous answers. Human grading is also used by in “SciEx” [30], where experts evaluate the LLM’s results. As acknowledged by Dinh et al. [30], human grading is not feasible for large-scale benchmarking due to its time and resource inefficiency.

2.5.1 CritPt

Presented by Zhu et al., “CritPt” [80] introduces checkpoints which allow to identify a model’s progress on a question. This provides a more in-depth view of the failure points of the evaluated system. The probed system is prompted with the question and generates an answer; afterwards, the required output format must be adopted to enable automatic evaluation. Zhu et al. describe using the checkpoints to guide the system towards the correct answer and also mention a mechanism which assists the tested system in adopting the required output format. [80]

There are three types of questions, which require either a numerical value, a symbolic expression, or executable python code. Composite answers are only graded as correct if all individual parts are correct. [80]

CritPt tests a system on 70 problems which are publicly available via their GitHub repository [26], the scores are averaged over five runs. The answers to the problem sets are not made public to prevent models from finding them, rendering the benchmark useless. Therefore, the models answers to the questions are submitted to an evaluation server, returning the final result of the benchmark. Users are limited to 10 submissions per 24 hours, in order to prevent leakage of the correct answers. For the same reason, not solutions or attempts of them should be publicised. [80, p. 10]

3 Design of the Physics Modelling Assistant

In order to easily enable future work on and with the Physics Modelling Assistant, the choice was made to only use services which are open source and free of charge.

3.1 Orchestration Framework: CrewAI

The decision was made not to design a new framework for orchestrating AIs in order to save time on this software development heavy task and instead focus on the Physic Modelling Assistant’s configurations. Derouiche, Brahmi, and Mazeni [29] discuss a number of frameworks varying in complexity and beginner friendliness. For this Thesis, the CrewAI [24] framework is used for its role focused style [29], consisting of Agents and Tasks [24] providing a level of abstraction from the technical details of orchestration. Every Agent can be configured individually, which includes using different LLMs for every Agent. Every Task has a description and specifications regarding the output and the output format, along with the Agent designated to it [25]. On execution of a Task, its instructions are merged with the assigned Agent’s, providing the input for the underlying LLM. An Agent’s behaviour is defined by its role, goal, and backstory. The role is a very short description comparable to a job title. The goal specifies the actions an Agent is supposed to perform similar to a job description. Details for the Agent’s behaviour are provided via the backstory argument, including preferences or advice on how to approach the Task. CrewAI allows to access any LLM, provided

the user presents a valid API key for the model. The LLM can be assigned parameters including the discussed Temperature (subsubsection 2.2.3) and top_p (subsubsection 2.2.5), the number of tokens to be used, and others. Tools can be assigned to both Agents and Tasks, with CrewAI providing ready to use tools, but also allowing to add custom tools [19].

The organisation of multiple CrewAI Agents and Tasks is managed by the Crew [20]. As depicted in Figure 1, it is responsible for initializing and validating the Agent's and Task's configurations before launching the system. As stated by Derouiche, Brahmi, and Mazeni [29], CrewAI's orchestration combines elements from the ACP, A2A and ANP (see subsubsection 2.4.2). The CrewAI documentation [24] and the source code available on GitHub [23] show a sequential and a hierarchical execution approach. The sequential execution (see Figure 1) runs the Tasks in the order they are defined in the source code, thereby forming a sequential Pipeline (see paragraph 2.4.1). This can be modified into a Parallel Fan-Out with Merge (see paragraph 2.4.1) by, allowing certain Tasks to run at the the same time, which CrewAI calls asynchronous [22]. Merging is handled by specific Task, waiting for all parallel Tasks to be done before starting its work (see Flows [21]). The hierarchical execution style (see Figure 1) of a CrewAI Crew enables a Hierarchical Supervisor-Worker architecture (see paragraph 2.4.1) by delegating Tasks, or aspects of them, to Subtasks. Implementing a Reflexive Self-Correcting Loop (see paragraph 2.4.1) with CrewAI can be achieved by making use of Flows [21], which allow for conditional calls and the coupling of multiple Crews.

In order to achieve a deterministic workflow and allowing to easily make adjustments to the setup, the Sequential Pipeline architecture (paragraph 2.4.1) is used for the Physics Modelling Assistant.

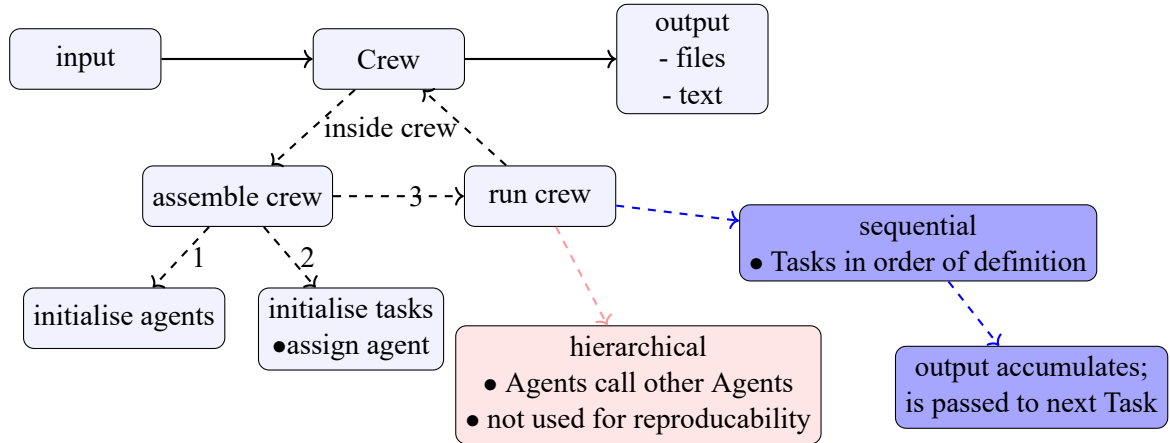


Figure 1: The main steps executed during a run of a CrewAI Crew. The Crew receives an input containing various parameters specifying the Crew's behaviour. This includes a message stating what the Crew should work on, as well as parameters shaping the LLM's behaviour (Temperature subsubsection 2.2.3, top_p subsubsection 2.2.5), and the Agent's behaviour (planning before execution, how many times to rerun itself [16]). The Crew is assembled by initializing all of its Agents and Tasks, before mapping the Agents to their corresponding Tasks. If there is a Task without an Agent, the Crew produces an error and does not proceed with the execution. After the Crew is assembled correctly, it can run in a hierarchical or a sequential style. For the hierarchical execution one main Agent calls other (Sub-)Agents, which can call further (Sub-)Agents themselves, to aid them in accomplishing their given Task (see subsection 3.1). During the sequential execution, the Agents, respectively the Tasks with their Agents, are executed in the order they are defined in the source code (see subsection 3.1).

3.2 LLMs Used

In order to use LLMs effectively in an automated pipeline, a provider offering a program access to its LLMs is needed. For this Thesis, the “Scalable AI Accellator (SAIA)” [31] hosted by the “Gesellschaft für wissenschaftliche Datenverarbeitung mbH Göttingen” (GWDG) [44] is used, offering limited but free access to a selection of open weight LLMs. The models are accessed via an API key issued by the GWDG. A list of the available LLMs is provided on the SAIA webpage [28], but can also be obtained via a request to the server hosting the models. To identify to most capable of the available models, individual benchmarks contributing to the *LLM Leaderboard* from Artificial Analysis are considered [2]. The AA-LCR [3] tests the long context reasoning, the GPQA Diamond [5] tests for scientific reasoning capabilities, SciCode [7] evaluates the models capabilities in science and coding, CritPt [4] (see subsection 2.5.1) probes the physics based reasoning abilities of the LLMs. The performances of all available models across the mentioned benchmarks are displayed in Table 1. They conclude qwen3.6 27B (27 billion parameters) to be the best model for long context reasoning, while qwen3.5 397B A17B (397 billion parameters in total, 17 billion active at a time) performs best in scientific reasoning tested by GPQA Diamond. glm-4.7 (358 billion parameters [43]) achieves the best result in science and coding. For the CritPt benchmark both glm-4.7 and qwen3.5 397B A17B achieve the best score among the tested models, with both of them scoring at only 1.7 %, indicating the difficulty of the CritPt benchmarking questions.

Table 1: Benchmarking results for the LLMs provided by SAIA [31]. The benchmarking results are taken from Artificial Analysis [6]. The chosen benchmarks are: AA-LCR evaluating long context reasoning, GPQA Diamond evaluating scientific reasoning, SciCode evaluating coding and science capabilities, CritPt evaluating physics reasoning

model	AA-LCR in %	GPQA Diamond in %	SciCode in %	CritPt in %
apertus 70B Instruct	0	27	6	0
deepSeek V4 Flash	33	72	37	0
devstral 2	30	53	29	0
gemma 4 31B	62	86	43	1
glm-4.7	64	86	45	1.7
llama 3.1 8B	16	26	13	0
mistral Medium 3.5	61	75	40	0
gpt-oss-120B (high)	51	78	39	1
qwen3 30B A3B 2507	59	71	33	0
qwen3 coder next	40	74	32	0
qwen3 omni 30B A3B	0	73	31	0
qwen3.5 122B A10B	67	86	42	1
qwen3.5 397B A17B	66	89	42	1.7
qwen3.6 27B	69	84	40	1
qwen3.6 35B A3B	64	84	36	0

3.3 Tools Used

To enhance the Physics Modelling Assistant’s capabilities, certain Agents are equipped with tools (see subsection 2.3.2). Each tool provides an object specifying the expected input at its constraints, both are visible to the Agent using the tool.

3.3.1 arxiv_paper_tool with Exponential Backoff

CrewAI already provides the “arxiv_Paper_Tool” [18], allowing Agents to access the arXiv-API [8] for finding papers relevant to the question at hand. During the development process of the Physics Modelling Assistant, the CrewAI tool’s requests repeatedly kept getting denied, resulting in no papers to be found and downloaded from arXiv. As listed in the arXiv-API’s Terms of Use, rate limits are enforced, blocking clients from sending many requests in a short amount of time. Furthermore, web servers deny repeated request from the same client for the same service [75]. Since the CrewAI arxiv_paper_tool [18] does not implement any strategy for waiting, its requests are eventually blocked by the arXiv-API servers. As a consequence an extending to the arxiv_paper_tool was made by implementing an exponential back-off strategy, as it is common for modern server requests [75]. Additionally to the exponential back-off, the modified tool, called arxiv_paper_tool_exponential_backoff, adds a small random wait, which is common to prevent two clients requesting the same service from waiting for the same amount of time blocking each others requests repeatedly [75]. The tool finds a specified number of papers and downloads them to a designated directory.

3.3.2 pdf_reader

In order to access the papers downloaded from arXiv, a PDF reader tool was developed. It reads all PDF files from the specified directory, allowing the invoking Agent’s LLM to see the contents of the documents. In theory this allows to read any number and length of sources from the directory. In reality every LLM has a context-window, limiting the amount of tokens it can handle. When the sum of the tokens of the read documents exceeds the context-window, the Agent fails its Task. Among the SAIA models [28], deepSeek V4 Flash offers the largest context-window, supporting up to 1 million tokens, while qwen3.6 35B A3B provides the second-largest context-window at 262k tokens.

Since the tool only allows reading files from a hard-coded subdirectory, it is deemed secure to be used outside of a sandbox environment (see subsection 2.3.2).

3.3.3 dimensional_analysis

During the testing of the system, formulas from source papers were stated correctly, but assigning correct units to the individual quantities proved to be difficult for the responsible Agent. Mistakes in the unit assumptions lead to unphysical results or impossible units for the outcomes of the numerical implementation. To verify the consistency for all formulas used, the dimensional_analysis tool was created. It is based on the sympy library and allows an Agent to check the units it proposes. The tool takes an equation written as a string, a dictionary containing the used quantities and their proposed units, as well as a list of all units that are used. The units are substituted in for their respective quantities and both sides of the formula are simplified separately. Numerical values are not considered, since the aim is simply to achieve unit consistency. Lastly, the left side of the formula is divided by the right side, giving the unit correction needed on the right side for the formula to be consistent. Ideally, this value is 1, indicating no correction is needed. If it is not 1, the Agent is supposed to interpret the unit mismatch correctly, by adjusting the assumed units for the used quantities. Another option for the Agent is to modify the formula, which can be done by inserting physical constants. This allows to keep real world units while implementing a formula, which uses a different unit system.

3.3.4 Code Execution

As discussed by Marchand et al. [55], a secure Sandbox is complex to design, but necessary, if LLM generated code should be executed directly. As stated in subsection 2.3.2, the weak points of Sandboxes can be exploited by capable LLMs, which is why the Physics Modelling Assistant does not allow code execution.

3.4 Experimental Setup for the Pipeline

The final system is designed to mirror a simplified research workflow consisting of finding relevant information, developing a model based on the knowledge acquired, and implementing the model to get an idea of what it predicts. In order to properly answer the questions from benchmarks (see subsection 2.5), the final Task is responsible for providing the output in the benchmark's required format.

A simplified view of the Physics Modelling Assistant's steps is shown in Figure 2, each step is performed by one Task and its corresponding Agent.

3.4.1 Input

The Physics Modelling Assistant receives the problem description, a string called **aim**, and instructions for the required format, also as a string. Additional parameters can be chosen via an interface, but defaults are set. The output directory specifies where the Tasks dump their output files; the directory's name consists of the question number and a unique identifier, ensuring previous outputs are not overwritten. The Temperature (see subsection 2.2.3) and the **top_p** (see subsection 2.2.5) parameters can be specified to shape the LLM's output. The **max_tokens** parameter limits the number of tokens a LLM produces. It is set to prevent the context-window from being exceeded, which would cause the Agent to fail its Task.

Each Agent can perform **reasoning** before it works on its Task. Reasoning implies that an Agent creates a plan for how to approach the Task before working on it [17] (see subsection 2.3.3). The maximum number of reasoning tries can be specified; otherwise the Agent retries to create a plan until an attempt is deemed sufficient [17]. An Agent can also rerun itself to improve its output if it deems this necessary. The maximum number of iterations is set through the **max_iter** parameter [17], the default is 20.

The directory for the Tasks output files and the PDF downloads can be specified in the input as well.

3.4.2 General Parameter Choices

The Temperature and the **top_p** parameter are the same for all Agent, based on the recommendations from the GWDG webpage [28]. This aligns with the statements from subsection 2.2.1, describing the decoding of a LLM's output as an optimization problem. For glm-4.7 and qwen3.6 27B the recommended parameters are: **Temperature** = 1.0 and **top_p** = 0.95. For qwen3.5 397B the recommendation is: **Temperature** = 0.6 and **top_p** = 0.95. For the reasons discussed in subsection 3.4.4, the qwen3.5 397B model is not used, which allows to set the Temperature and **top_p** globally.

The maximum number of output tokens is set to 50k [17] to ensure the context-window is not exceeded. This is especially relevant for the Agent extracting information from the sources.

Each Agent is allowed up to three iterations (**max_iter** = 3) to refine its result if it deems this necessary [17]. Reiterations allows a single Agent's result to be improved, if its output is of bad quality. This can happen due to the complexity of the Task, the inherent randomness of LLMs, or temporary problems with accessing servers. Furthermore, retries can mitigate the influence of infrequent failures like the context-window being exceeded.

In order to balance the performance gains from reasoning (subsubsection 2.3.3) with the need to limit excessive retries (subsubsection 3.4.1), the **max_reasoning_attempts** parameter is set to 3 globally.

The Physics Modelling Agent does not use specific memory (subsubsection 2.4.3) but relies on CrewAI passing the outputs of previous Agents to later ones via a shared output object (see [23, line 1610]).

3.4.3 Obtaining Information

Source Gathering In order to get sufficient information for building a model, the arXiv-API [8] is accessed via the modified `arxiv_paper_tool` (see subsubsection 3.3.1) and the found papers are downloaded. The corresponding **source gatherer** Agent is based on the qwen3.6 27B LLM and is tasked to use its tool for finding papers related to the aim. The **gathering task** specifies that three to six papers should be found, more papers bare the risk of exceeding the context-window of the used LLMs. The Task further specifies which information should be listed for each source: arXiv ID, author, data, URL and a short summary. This content is written into a markdown report, allowing a human to inspect the sources found.

This Agent does not use reasoning because its Task is deemed to not need a multi step approach. All other parameters are kept at the global settings discussed in subsubsection 3.4.2.

Because of the issues with the arXiv-API rate limits [8], this step is packaged into its own Crew, allowing it to be executed independently from all later steps. This way changes or reruns intended to result in different behaviour of later steps can be made without rerunning the arXiv request again. For the measurements of the Physics Modelling Assistants performance in this Thesis, the source gathering is run again on every retry.

Information Extraction The source finding Task saves its papers to the specified output directory. By knowing the directory's location, the **information extractor** Agent can use its `pdf_reader` (see subsubsection 3.3.2) to read the content of the files in that directory. The qwen3.6 27B LLM is used for its superior long context reasoning score (see Table 1). Its context-window of 262k tokens is the second largest only to deepSeek V4 Flash's 1M [31] [28]. Since deepSeek V4 Flash achieves less than half of qwen3.6 27B's score in long context reasoning Table 1, the latter is used.

The **information extractor** is instructed to find the information needed to build a model that accomplishes what the aim asks for. It should present its findings with scientific citation and its backstory emphasises that the Agent should stay true to the source material it has available. The corresponding **extraction task** specifies that the output should be in markdown format, enforcing an unambiguous and human readable output for later Agents and inspecting the Tasks markdown output file.

The parameters are kept at the global settings as described in subsubsection 3.4.2

3.4.4 Building the Model

Modelling For building the model which is at the heart of the Physics Modelling Assistant, the LLM strongest in scientific and specifically physics reasoning should be used. Therefore, the first choice was qwen3.5 397B A17B performing the best at scientific reasoning (GPQA Diamond [65]) and joined best at CritPt [80], as shown in Table 1. During the testing phase, requests to this model kept failing with an answer not being provided by the server before a timeout was reached. For that reason the glm-4.7 LLM is chosen instead, for it being the joined second best model at scientific reasoning (GPQA Diamond [65]) and joined best model at CritPt [80] (see Table 1). The **modeller** Agent is prompted to provide a mathematical description for a model which accomplishes the aim. It is further instructed to rely on the information provided by the previous Task (extraction Task), provide a proper explanation for the steps taken, and not to write any code. The last instruction targets the Agent’s previous behaviour of occasionally writing code which already implements the model. This is not wanted due to potential modifications needed for the model to be sufficient.

The **modelling task** defines the expected output to be a complete mathematical description with a textual explanation. The requested format is once again markdown ensuring human readability of the generated output file. The default parameters setting described in subsection 3.4.2 are adopted.

Unit Check The **unit checker** Agent accounts for custom parameter definitions and other conventions preventing a straight forward adoption of the model and its formulas. It uses the glm-4.7 LLM and its goal specifies which steps to take. In the beginning the units of all the quantities should be determined and based on these assumptions, the `dimensional_analysis` tool (see subsection 3.3.3) is used to identify potential inconsistencies in the units assumed. Based on the tool’s results, the Agent is expected to correct its unit assumptions or formulas if deemed necessary.

The **unit checking task** specifies that the quantities of the units and the output of the dimensional analysis tool should be displayed in the markdown output file. The parameter configuration follows the default (see subsection 3.4.2).

Parameters Suggestion In order to obtain a meaningful simulation, physically realistic starting conditions are required. They are provided by the **parameter suggester**, which has the goal of finding typical starting parameters for the model simulation, and stating how these suggestions were derived. The **parameter suggestion task** highlights that explanations for the choice of parameters should be included in the markdown output file. The input parameters are unchanged compared to the default and the glm-4.7 model is used for the reasons discussed in paragraph 3.4.4.

3.4.5 Implementing the Model

Physics Simulation In order to capture the suggested model accurately, the **simulation physician** Agent is run with the goal of implementing the formulas of the model precisely, without making changes to them. The Agent is based on the glm-4.7 LLM, since it performs the best in coding and science among the available models (see Table 1). As specified by the **physician simulation task**, the implementation is supposed to use the units determined by the unit checking task. To get human readable and interpretable results, the task is requested to create graphics if it deems this appropriate. The output is a Python file which should only

contain executable Python code, all explanations should be in comments. The parameters are kept the same as before.

Simulation Correction According to its specifications the physics simulation task is meant to capture the models logic into working code, it is not asked to prioritize producing error free code. The correction of bugs is done by the **simulation correcter**, which is told not change the implemented formulas, but to correct coding mistakes and improve the codes efficiency and style. The corresponding **simulation correction task** defines the expected output to be working python code. The parameter configurations and LLM are unchanged compared to the simulation physician in paragraph 3.4.5.

3.4.6 Giving a Final Answer

Benchmarks like CritPt expect a comprehensive answer (see subsection 2.5) to their questions, for that reason the **final outpiter** Agent is added to the pipeline. Its goal is to give a final answer and obey the format requested by the input question. Adapting the required format is crucial for CritPt’s grading process [80]. The **final output task** specifies the output is supposed to be a string and produces a markdown output file. The glm-4.7 model is used for its performance on the CritPt benchmark seen Table 1, and the issues with qwen3.5 397B A17B discussed in paragraph 3.4.4.

3.4.7 Output

The output of the Physics Modelling Assistant is an object containing a string with the last Task’s result as the answer to the benchmark’s questions. Output files generated by each Task can be inspected by a human; the Python scripts are not directly executable since they contain triple backticks at the beginning and end. Presumably, this is done to mark the section in between as code, which is not accurate Python syntax. Removing or commenting the backticks results in the scripts becoming executable, apart from possible package installations.

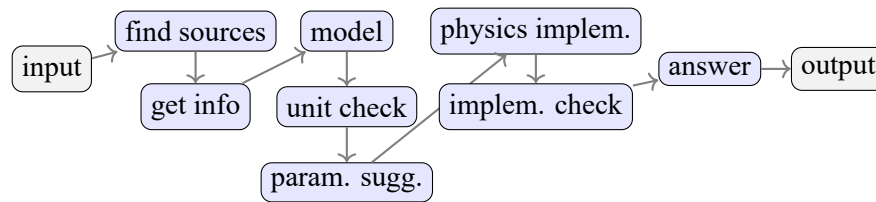


Figure 2: This figure provides an overview of all the Tasks and their order of execution, as performed by the Physics Modelling Assistant. The system receives an input string which asks a question or formulates a modelling request. It finds papers related to the aim and downloads them from arXiv; the papers are read and relevant information is extracted. A model is build based on the extracted information; the model’s formulas are checked for unit consistency; realistic starting parameters are suggested for the simulation. Lastly, the model is implemented into Python code; the code is checked and corrected if necessary. The last Task provides the final answer to the given input. Each Task dumps a file stating the work it did.

3.4.8 Connecting to CritPt's Benchmark

CritPt is based on the `inspec_ai` package, which allows to register a model by implementing the ModelAPI interface. This is achieved by the `PMAModelAPI` class, implementing the required `generate` method. This method receives the description of a CritPt challenge, but not its number or the required output format. This does not align with the statement from subsection 2.5.1, suggesting the system described by Zhu et al. [80] was modified compared to the one publicly available on GitHub [26]. The same is true for CritPt guiding the system towards the correct result via checkpoints. The 70 publicly available challenges do not include any checkpoints and the GitHub repository does not mention a way to obtain them `autocitecritptGit`.

The `PMAModelAPI` instance is programmed to identify the challenge based on the problem description it receives, obtaining the challenge number and the required output format.

Because of conflicting dependencies of the `inspect_ai` and `crewai` packages requiring incompatible versions of the `tiktoken` and `pydantic` packages, the `PMAModelAPI` instance runs the Physics Modelling Assistant as a subprocess in a different virtual environment. It passes the challenge description and output format to the subprocess, along with the parameters the Physics Modelling Assistant exposes (see subsection 3.4.2; `Temperature`, `top_p`, `max_tokens`, `max_iter`, `reasoning` (True or False), `max_reasoning_attempts`). The challenge number along with a unique identifier and a file path are passed to the subprocess as well. This allows the processes to communicate via reading and writing to a file specified by the location and the identifier. This was implemented to work around the problem of extracting the Physics Modelling Assistant's output from the standard output of the subprocess.

The subprocess consists of two layers. The top layer handles the communication with the super-process and starts the bottom layer, which is the Physics Modelling Assistant. The top layer handles the communication with the super-process. The Physics Modelling Assistant runs described in subsection 3.4 and returns its final output to the top layer. There, the result is written into the file specified by the super-process, which waits for the subprocess to finish its execution, before attempting to read from the file.

The generated results are stored in JSON-files and can be send to the evaluation server via a script provided by CritPt. Since Artificial Analysis [6] has integrated CritPt, a valid API key from Artificial Analysis [6] is needed for the evaluation server. It can be obtained via Artificial Analysis and is free of charge. For the evaluation requests to be approved, the account connected to the submitting API key needs to be marked as a trusted research account by Artificial Analysis [45].

3.5 Investigating the Physics Modelling Assistant's Performance

3.5.1 An Example Run

To provide inside into the Physics Modelling Assistant's work, the system is given an example modelling request. It is not taken from the CritPt benchmark, since no solutions are attempts of them should be published (see subsection 2.5). The following aim is chosen:

“Calculate the Edelstein effect for a Rashba fermion (at the Gamma point of the Brillouin zone). Compute the magnetization magnitude and direction of different directions and magnitudes of the applied electric field. Consider how the result depends on relevant parameters of the model (e.g. chirality, Fermi velocity, spin-orbit coupling strength) and make explicit graphics.” The Physics Modelling Assistant's work is presented for

a selected challenge from the benchmark. It is meant to provide insight into the system and the outputs it produces, which are meant to assist in physics modelling.

The final output is specified to be “simple markdown format”. The parameters are the defaults described in subsection 3.4.2.

3.5.2 Pipeline’s Performance Compared to Individual Models

The Physics Modelling Assistant’s performance is measured and compared to the results of the individual LLM’s performances stated in Table 1. The Pipelines performance in the described setup (see subsection 3.4) is used as a reference for other changes.

3.5.3 The Relevance of the Available Sources

The influence of the sources available to the pipeline is measured.

The pipeline runs with the deepSeek V4 Flash LLM used for the information extraction (see paragraph 3.4.3). Its performance is compared to the reference which uses qwen3.6 27B.

The Physic Modelling Assistant’s performance is measured when it has no external sources available. The performance is compared to the the reference setup.

3.5.4 The Effect of Iterations of the Agents

All Agents of the Physics Modelling Assistant receive $\text{max_iter} = 1$, which means they can not reiterate. The result are compared with the default setup, which allows for 3 iterations.

4 Results and Discussion

4.1 Results of the Example Run from the Reference Setup

Found Sources A total of six papers are downloaded from arXiv. These are:

1. “Edelstein Effect in Isotropic and Anisotropic Rashba Models” [38]
2. “Out-of-plane Edelstein effects: Electric field induced magnetization in wave magnets” [36]
3. “Acoustic Rashba–Edelstein effect” [37]
4. “Spin accumulation at nonmagnetic interface induced by direct Rashba–Edelstein effect” [10]
5. “Electric-Dipole Spin Resonances” [64]
6. “Out-of-Plane Spin Polarization from In-Plane Electric and Magnetic Fields” [35]

The markdown report includes the following information for each paper: arXiv ID, authors, date, URL to the paper on arXiv, and a short summary. For the full report visit [63].

The Task works as expected, providing all the requested information and staying within the limit of downloading a maximum of six papers. All sources directly relate to the Edelstein effect and discuss aspects of it, which are relevant to the modelling request (see subsection 3.5.1)

Extracted Information The report on this Task lists all of the downloaded papers along the following additions, which were not downloaded:

7. “Properties of a 2D electron gas with lifted spectral degeneracy” [13]
8. “Spin polarization of conduction electrons induced by electric current in two-dimensional asymmetric electron systems” [33]
9. “Current-induced spin-orbit torques” [39]

The full report (see [56]) lists and explains a variety of formulas, starting with the Rashba Hamiltonian:

Hamiltonian found

The **Rashba Hamiltonian** for a 2DEG at the Γ point is [Bychkov & Rashba, *JETP Lett.* 39, 78 (1984)]:

$$\hat{H} = \frac{p^2}{2m} + \alpha \hat{z} \cdot (\mathbf{p} \times \boldsymbol{\sigma})$$

where:

- m is the effective carrier mass,
- α is the Rashba spin–orbit coupling (SOC) strength,
- $\mathbf{p} = \hbar \mathbf{k}$ is the momentum,
- $\boldsymbol{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$ are the Pauli matrices,
- $\hat{z}$ is the direction perpendicular to the 2D plane.

Most contents are used from and attributed to the work of Gaiardoni et al. [38], these include the *energy dispersion*, the *spin eigenstates*, and the *Fermi wavenumbers* split into high density regime (HDR) and low density regime (LDR). The formula for the magnetization (**total spin density**) is stated from [38] and [36]. For the Magnetization, the following analytical results for the isotropic model are provided:

Analytical magnetization

4.1 Magnetization in the High-Density Regime (HDR)

For an electric field $\mathbf{E} = E_x \hat{x}$, the spin density along $\hat{y}$ in the HDR (both bands occupied) is [Gaiardoni et al., arXiv:2503.20712 (2025)]:

$$M_y = \frac{\mu_B |e| \tau}{2\pi} m \alpha [\hat{z} \times \mathbf{E}]_y$$

Key result: In the HDR, the magnetization is **constant, independent of the Fermi energy** E_F , and proportional to the Rashba SOC parameter α and the effective mass m .

4.2 Magnetization in the Low-Density Regime (LDR)

In the LDR (only the lower band occupied) [Gaiardoni et al., arXiv:2503.20712 (2025)]:

$$M_y = \frac{\mu_B |e| \tau}{2\pi} \sqrt{m^2 \alpha^2 + 2m E_F} [\hat{z} \times \mathbf{E}]_y$$

For small E_F near the band crossing, this can be expanded as:

$$M_y \approx \frac{\mu_B |e| \tau}{2\pi} \left(\alpha m + \frac{1}{2} \frac{E_F}{\alpha} \right) [\hat{z} \times \mathbf{E}]_y$$

The spin density **increases linearly with the Fermi energy**. In the limit $\alpha \rightarrow 0$, the second term vanishes because E_F has a quadratic dependence on α .

Further information include the **Anisotropic Rashba Model**, the **Out-of-Plane Edelstein Effect**, the **Acoustic Edelstein Effect**, and the **Effect of a Magnetic Field**. The dependencies on the model's parameters is discussed with the following takeaways:

Edelstein susceptibility $\chi_{xy} \propto \alpha$ in HDR

$\chi_{xy}(E_F) = \text{const.}$ in HDR

$\chi_{xy}(E_F) \propto E_F$ linear increase with E_F in LDR

sign of α determines spin orientation on the Fermi surface

Lastly, the Task provides guidelines for a numerical implementation along with code snippets and the expected graphics.

The output draws its information from the papers downloaded from arXiv (see paragraph 4.1) and also provides proper citation for its statements. As shown in paragraph 4.1 and paragraph 4.1, the provided information are relevant to the problem at hand, even providing formulas that can be implemented. The sections about the Out-of-Plane Edelstein Effect and the Acoustic Edelstein Effect discuss more specialized cases not integral to the Edelstein Effect the input asks for. It appears that the inclusion of these information is heavily influenced by the downloaded papers [36], [37], and [35], which discuss these particular systems. The inclusion of implementation instructions is not expected. It could be problematic since the formulas not checked for dimensional consistency at this stage. Therefore, possibly incorrect implementation ideas are listed and can influence later Tasks to adopt them.

The Model Since the modelling Task reiterates the extracted information as stated in paragraph 4.1, the output is omitted here, but can be accessed via GitHub [57]. If the file appears not to be displayed correctly on GitHub, it might be due to the Markdown viewer. During the work on this Thesis, *Markdown Preview Enhanced* [74] was used, providing a clean human readable look except for incorrect Markdown code.

The modelling step is generally expected to use the information as a foundations and derive new formulas needed to provide an answer to the input. Contrary to the expectation, source code for the implementation was stated, matching the one from the information extraction paragraph 4.1. The provided information being restated point to the Task and its Agent either relying too much on the papers found, or deeming the information as sufficient for the modelling request. Furthermore, the request is comparably broad, not asking for a specific properties but general parameter dependencies. For the CritPt challenges the modelling step shows the intended behaviour, deriving new formulas not stated in the extracted information and integrating the explicit problem from the challenge description.

Unit Check Key formulas are checked for dimensional consistency regarding their units. The first step analysis the Rashba Hamiltonian as it is stated in paragraph 4.1:

Unit analysis of the Rashba Hamiltonian

Formula

$$\hat{H} = \frac{p^2}{2m} + \alpha \hat{z} \cdot (\mathbf{p} \times \boldsymbol{\sigma})$$

Quantities and Units - H (Hamiltonian): Energy $\rightarrow [E]$ (Joules) - p (Momentum): $[p] = [M][L][T]^{-1}$ - m (Mass): $[m] = [M]$ - α (Rashba parameter): $[\alpha] = ?$ - $\boldsymbol{\sigma}$ (Pauli matrices): Dimensionless

Tool Input

equation: $H = p^2 / (2*m) + \alpha * (p)$ dimensions: "H": "energy", "p": "momentum", "m": "mass", "alpha": "alpha_dimension" unitList: energy, momentum, mass, alpha_dimension

Tool Output

$2*energy*mass/(momentum*(2*alpha_dimension*mass + momentum))$

Analysis The first term $\frac{p^2}{2m}$ has units $\frac{([M][L][T]^{-1})^2}{[M]} = [M][L]^2[T]^{-2} = [E]$. The tool output shows that for consistency, *alpha_dimension* must satisfy *energy = momentum * alpha_dimension*. Since $[E] = [M][L]^2[T]^{-2}$ and $[p] = [M][L][T]^{-1}$, we must have:

$$[\alpha] = \frac{[E]}{[p]} = \frac{[M][L]^2[T]^{-2}}{[M][L][T]^{-1}} = [L][T]^{-1}$$

Conclusion: The Rashba parameter α has units of **velocity** (m/s)

The energy dispersion is deemed dimensionally consistent by the tool output and no correction is applied.

Unit correction for the HDR magnetization

Formula

$$M_y = \frac{\mu_B |e| \tau}{2\pi} m \alpha [\hat{z} \times \mathbf{E}]_y$$

For dimensional analysis, we examine the product $M_y \propto \mu_B e \tau m \alpha E$.

Quantities and Units

- M_y (Magnetization/Spin Density): Magnetic moment per unit volume.
- In SI (Volume): $[M] = [I][L]^2 \text{ (Am}^2\text{)} / [L]^3 = [I][L]^{-1} \text{ (A/m)}$.
- In 2D (Area): $[M] = [I] \text{ (Amperes)} / [L]^2 = [I][L]^{-2} \text{ (A/m}^2\text{)}$.
- In microscopic units (just density): $[M] \propto [\hbar]^{-1} [L]^{-2}$.
- μ_B (Bohr magneton): $[\mu_B] = [I][L]^2 \text{ (Am}^2 \text{ or J/T)}$.
- e (Elementary charge): $[e] = [I][T]$.
- τ (Relaxation time): $[\tau] = [T]$.
- m (Mass): $[m] = [M]$.
- α (Velocity): $[\alpha] = [L][T]^{-1}$.
- E (Electric field): $[E] = [M][L]^3 [T]^{-3} [I]^{-1} \text{ (V/m)}$.

Tool Input

equation: $M_y = \mu_B * e * \tau * m * \alpha * E$ dimensions: "M_y": "magnetization", "mu_B": "magnetic_moment", "e": "charge", "tau": "time", "m": "mass", "alpha": "velocity", "E": "electric_field" unitList: magnetization, magnetic_moment, charge, time, mass, velocity, electric_field

Tool Output

magnetization*exp(-1)/(charge*magnetic_moment*mass*time*velocity*electric_field)

...omitting a long analysis which is part of the original output...

Conclusion on Formula Consistency: The formula is dimensionally consistent **if and only if** the system of units uses $\hbar = 1$ (natural units) or the parameters m and α are defined in units that include $\hbar$ (common in computational materials science, e.g., α in $eV \cdot \text{\AA}$ and m in $eV^{-1} \text{\AA}^{-2} \cdot \hbar^2$). However, in standard SI units, the formula is missing a factor of $\hbar^{-2}$.

Correction for Standard Units:

$$M_y = \frac{\mu_B |e| \tau}{2\pi \hbar^2} m \alpha [\hat{z} \times \mathbf{E}]_y$$

For the LDR magnetization the same correction is applied. Additionally, formulas for the anisotropic susceptibility and the Out-of-Plane susceptibility are analysed. The output does not include any tool calls and is unable to identify a unit system for both formulas. Lastly, the findings are summarised, restating the $\frac{1}{\hbar}$ correction for both HDR and LDR magnetization. The full report can be inspected at [58]

This task correctly identifies the units of the Rashba coupling strength α to be a velocity for the given Hamiltonian (see paragraph 4.1). The corrections for the magnetization achieve unit consistency, as stated by the output. This is checked by hand to verify the result: The used properties and their units are:

- M_y is the magnetization. As stated in the extracted information in paragraph 4.1, it is the total spin density (Am^2) per unit volume. Since Gaiardoni et al. [38] discuss a 2D system, the unit volume is in units of m^{-2}
- μ_B is the Bohr magneton. It can be expressed in JT^{-1}
- e is the charge with the unit of As
- τ is the transport time expressed in s
- m is the effective mass in kg
- α is the Rashba coupling strength. As stated in paragraph 4.1, it is unit of velocity, ms^{-1}
- $\hat{z}$ is a dimensionless direction
- $\mathbf{E}$ is the applied electric field. It can be expressed in Vm^{-1}
- $\hbar$ is the Planck constant with SI units of Js

The relevant unit conversions are:

$$1\text{T} = 1 \frac{\text{Vs}}{\text{m}^2} \quad (4.1.1)$$

$$1\text{J} = 1\text{kg} \frac{\text{m}^2}{\text{s}^2} \quad (4.1.2)$$

Using these units leads to the following:

$$[M_y] \stackrel{!}{=} \left[\mu_B \|e\| \tau m \alpha (\hat{z} \times \mathbf{E})_y \frac{1}{2\pi\hbar^2} \right]$$

$$\text{A} \stackrel{!}{=} \frac{\text{J}}{\text{T}} \cdot \text{As} \cdot \text{s} \cdot \text{kg} \cdot \frac{\text{m}}{\text{s}} \cdot \frac{\text{V}}{\text{m}} \cdot \frac{1}{\text{J}^2 \text{s}^2}$$

now use Equation 4.1.1:

$$(4.1.3)$$

$$\text{A} \stackrel{!}{=} \frac{\text{Jm}^2}{\text{Vs}} \cdot \text{As} \cdot \text{s} \cdot \text{kg} \cdot \frac{\text{m}}{\text{s}} \cdot \frac{\text{V}}{\text{m}} \cdot \frac{1}{\text{J}^2 \text{s}^2}$$

$$\text{A} \stackrel{!}{=} \frac{\text{Jm}^2}{\text{Vs}} \cdot \text{As} \cdot \text{s} \cdot \text{kg} \cdot \frac{\text{m}}{\text{s}} \cdot \frac{\text{V}}{\text{m}} \cdot \frac{1}{\text{J}^2 \text{s}^2}$$

$$\text{A} \stackrel{!}{=} \text{A} \frac{\text{kgm}^3 \text{s}^2}{\text{Js}^4 \text{m}}$$

$$\text{A} \stackrel{!}{=} \text{A} \frac{1}{\text{J}} \cdot \text{kg} \frac{\text{m}^2}{\text{s}^2}$$

now use Equation 4.1.2:

$$\text{A} \stackrel{!}{=} \text{A} \quad \text{verifies unit consistency} \quad (4.1.4)$$

If $[\alpha] = \text{eV}\text{\AA} \stackrel{!}{=} \text{Jm}$ as stated in “Edelstein Effect in Isotropic and Anisotropic Rashba Models” [38], and additional $\frac{1}{\hbar}$ correction is needed, resulting in $[\frac{\alpha}{\hbar}] \stackrel{!}{=} \frac{\text{Jm}}{\text{Js}} \stackrel{!}{=} \frac{\text{m}}{\text{s}}$.

The tool not being called for the LDR magnetization formula is likely due to the formula requiring the same correction factor as the HDR magnetization. For the anisotropic and Out-of-Plane susceptibility, the absence of the dimensional analysis tool being invoked might be due to the Agent’s inability to come up with a proper unit system in the first place.

Parameter Suggestions For each parameter the Task lists its range, a specific suggestion, and a justification for the range and suggestion are provided. The justification for the effective mass m and the Rashba coupling

strength α include a source backing the suggestion. There are two different unit suggestions for the effective mass, the first is to use $0.13 \frac{1}{\text{eV}\text{\AA}^2}$, the second one to use $9.11 \cdot 10^{-31} \text{kg}$. The final parameter suggestions is:

Parameter suggestions for the implementation

Parameter	Symbol	Value	Units	Physical Source
Effective Mass	m	$0.5 m_e$	kg	$m_e = 9.11 \times 10^{-31} \text{ kg}$. Effective masses in LAO/STO range from 0.5 to $3m_e$ [1].
Rashba Coupling	α	5.0×10^{-11}	eV·m	Corresponds to $\approx 50 \text{ meV}\cdot\text{\AA}$. This is a "strong" SOC regime observed in Bi/Ag or gated interfaces [2, 3].
Fermi Energy	E_F	60	meV	Typical carrier density $n \sim 10^{13} \text{ cm}^{-2}$. Places system in HDR ($E_F \gg E_R$).
Relaxation Time	τ	1.0×10^{-12}	s	1 picosecond. Corresponds to a mobility $\mu \approx 10 \text{ cm}^2/\text{V}\cdot\text{s}$ for $m \approx m_e$ [1].
Electric Field	E_x	1000	V/m	Linear response regime. Small enough to avoid Joule heating, large enough for signal.
g-factor	g	2.0	-	Free electron value.
Temperature	T	4.2	K	Low temperature limit (often used in theoretical/comparison papers).

Sources:

1. **Caprara, S. et al.** *Rep. Prog. Phys.* **79**, 096501 (2016). (Review of LAO/STO parameters: masses, mobilities, densities).
2. **Ast, C. R. et al.** *Phys. Rev. Lett.* **98**, 186807 (2007). (Giant Rashba splitting on Bi/Ag surface, $\alpha \sim 3 \text{ eV}\cdot\text{\AA}$).
3. **Ben Shalom, M. et al.** *Phys. Rev. Lett.* **105**, 206801 (2010). (Tunable α and E_F in LAO/STO).

The full report is accessible via GitHub [59]

With the exception of the Rashba coupling strength and the effective mass, the suggested parameters are in line with the previous findings. The suggestion for the effective mass m to be in $\frac{1}{\text{eV}\text{\AA}^2}$ is justified by working in atomic units. Contradicting the behaviour of the Task definition, the suggested unit for the Rashba coupling strength α of $\frac{\text{m}}{\text{s}}$ is not adopted, instead the suggested parameter is in eV·m. As discussed in Equation 4.1.3, this choice requires an additional correction of $\frac{1}{\hbar}$ to the formula for the magnetization. It appears to be that the Task relies on the parameters found other sources but is unable to adapt them to the corrected units.

Regarding the stated sources, the first one could not be found, suggesting it does not exist and therefore is a hallucination. The second and third sources are valid and can be found under [9] and [12].

Simulation The two simulation Tasks create two functionally identical files with only minor differences regarding comments. On execution the code calculates the Rashba effect induced change of the minimum energy E_R . It is compared to the Fermi energy to determine if the system is in the high density regime (HDR)

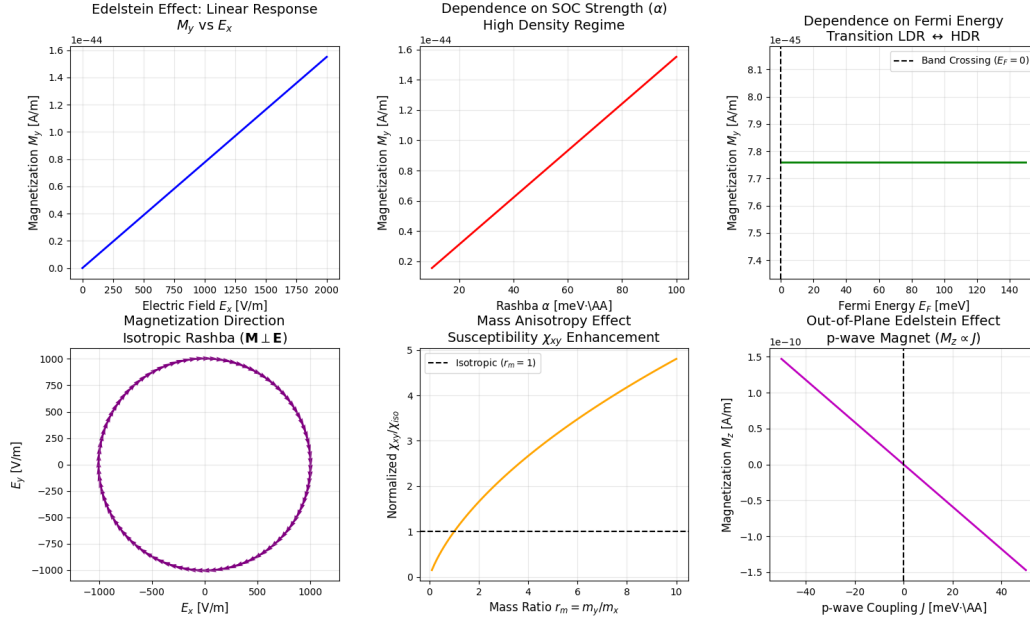


Figure 3: Output of the Physics Modelling Assistant's simulations after the correction where applied.

or low density regime (LDR), with the Fermi energy being higher than E_R indicating the HDR. The code's output is a graphic showing various dependencies for the Rashba effect. The unmodified code is available at [60], the corrected version at [61].

The code uses the Rashba coupling strength in J m, but does not apply the necessary correction of $\frac{1}{\hbar}$ (see Equation 4.1.3). Therefore, the initial results are off by this factor, giving a magnetization M_y in the range of $10^{-44} \frac{\text{A}}{\text{m}}$. The unit for the magnetization is also incorrect; as stated in paragraph 4.1, it should be A. To obtain more meaningful results, the following modifications are applied by hand:

- the magnetization formulas for both HDR and LDR got an additional $\frac{1}{\hbar}$
- the axis labels for the unit of the magnetization M_y were changed to Am/m^2
- the axis labels for the unit of the Rashba coupling strength α were changed fixed from $\text{eV} \cdot \text{\AA}$ to $\text{eV} \cdot \text{\AA}$

Applying these changes yields the following output graphic:

The simulation implements the model according to the units used for the suggested starting parameters (see paragraph 4.1) instead of the units stated by the paragraph 4.1. Therefore, manual corrections are required to obtain a more realistic result. Since the implementation Task produces error free code, the following correction Task does not need to make any fixes. However, the general coding style is not suited well for investigating the LDR, due to all plots being implemented statically, meaning they do not adapt to a single global change of the Rashba coupling strength or the Fermi energy.

Final Answer The final output mostly reiterates the extracted information (see paragraph 4.1). The full Markdown report is available at [62].

Since no specific output format is given and no concrete calculations are requested, the Task is not necessary to this run of the Physics Modelling Assistant.

4.1.1 Discussion of the Results from the Example Run

In its current configuration the Physics Modelling Assistant properly finds related sources from arXiv for the given input. It extracts relevant information and is able to apply unit corrections to extracted formulas. Failures occur when finding realistic starting parameters and implementing the extracted formulas, with the identified unit corrections not being considered. The final Task providing an answer to the input is only relevant if a concrete question is given, not for a broad modelling request.

Future modifications should address the system not acknowledging the units it proposes. The desired behaviour might be achieved by specifying that the responsible Task accepts the units stated by previous step.

Analytical Magnetization

Inhalt...

4.2 Evaluation of the Physics Modelling Assistant

4.2.1 Issues Encountered during the Generation

The most common reason for failures is the API rate limit of SAIA [28] being exceeded, resulting in a 429 error code. Rarely the context-window of a LLM is exceeded, resulting in the run to fail. Lastly, the adoption of the output format is inconsistent. As stated in subsection 3.4.7, the backticks and the work “python” needed to be removed, which is accomplished via an automated script. However, some results include additional descriptions not marked as comments before the output format is adapted, requiring further post generation correction to establish the proper format. On a few occasions, the output format is not adapted at all with only markdown like description being returned. If the post processing is unable to obtain the correct output format, the challenge is retried.

4.2.2 Benchmarking Results for the Default Setup

Three independent generation runs are conducted and evaluated. The CritPt evaluation returns the following accuracies: 2,86%, toBeObtained%, toBeObtained%.

Results are better or worse than individual models results? if Better: combining yields positive effect; if same: no positive effect => better to just use individual models

4.2.3 Benchmarking Results when Using DeepSeek V4 Flash

For the Crew using deepSeek V4 Flash instead of qwen3.6 27B for the information extraction Task (see paragraph 3.4.3), the following accuracies are achieved: toBeObtained%, toBeObtained%, toBeObtained%.

If Better, Worse, Same as default => longer context window superior, extraction capabilities more important than how much content is available, model choice not particularly important

4.2.4 Benchmarking Results without Accessing External Sources

For the Physics Modelling Assistant not using external information, the following accuracies are measured: toBeObtained%, toBeObtained%, toBeObtained%.

Worse: models benefit from information: either data leakage from CritPt or less powerful models do not have enough information available Same: answers to CritPt challenges not publicly available, as stated by developers.

4.3 General Discussion of Benchmarking Results

For the benchmarking asking for specific results, the parameter suggestion and the implementation do not contribute to the output in a meaningful way. Instead, the modelling Task executes the necessary derivations and calculations required by the challenge. This points to the Physics Modelling Assistant's design not being optimized for this benchmark. Instead, its setup is more suitable to work on a broader modelling request as stated in subsection 3.5.1.

which PMA setup performs best; does PMA outperform individual models

More runs needed for proper evaluation, CritPt uses average over 5 runs to evaluate models Therefore, most results are inconclusive and require more runs

Runs take very long time, between 15 min and hour. This is due to stronger models being used. Early test runs weaker models took less time. SAIA rate limits did not allow for generation to run continuously

5 Conclusion and Future Work

- does the PMA perform better than its individual models: ?

If yes: align with effect shown for physics-intern, improving performance for various models by up to 80 percent

If no: similar effects to physics- intern could not be shown.

Regardless: testing models used for Physics Intern on Physics Modelling Assistant and vice versa: This should give insight into the usefulness of the Physics Modelling Assistant's architecture.

Future research may include a variety of tests regarding the Physics Modelling Assistant's configurations. Instead of providing an arXiv search, a physics textbook is provided, giving access to a more foundational knowledge base instead of published research, which most often does not cover standard knowledge. Since a textbook most certainly exceeds the context-window of the LLMs considered so far, a different provider would be required for this research. Altering the maximum number of iterations and reasoning attempts can provide insight into the effect of these influenced processes. The general assumption is, that reasoning and reiterations improve performance, a structured research can provide statistical results for how the performance scales for both settings.

To use the Physics Modelling Assistant effectively in scientific work, improvements are necessary. The instructions regarding the model's implementation need to be refined, requesting a generalised result, which can be adapted to investigate different setups of the crafted model. For further improvements architectural changes enabling feed back loops, and dynamically assigning Tasks to specialized Agents might be required. These changes would adapt common architectures from other Agentic AI systems like the "physics-intern" [53] or "SciSciGPT" [67], which use feedback loops as integral part of their systems. Within the CrewAI framework, feedback loops can be implemented via Flows [21], utilizing required a significant expansion to the state of the Physics Modelling Assistant, with Flows expanding the Crew [23] based design. Changes to the discussed setup should generally be considered, regarding the redundancy of the second simulation implementation step. Dynamically deciding which Agents to call should also improve the adaptability of the system regarding different types of input, with a dynamical setup able to omit specific Agents not needed to answer a concrete challenge or implement a broad model based on existing information.

In its current state, the Physics Modelling Assistant is useful more as prototype than a helpful assistant for a human scientist. The design process and its result provide some insight into the design of Agentic AI systems, pointing to the importance of a feedback architecture as the results from Kulkarni and Kulkarni [51] suggest. I believe the simple design is a good starting point for systematically investigating the influence of parameters on systems combining multiple LLMs. By implementing a pipeline architecture it provides an interesting addition to the current landscape of Agentic AI frameworks for science, allowing future work to compare the different architectures in more detail.

Beyond the results presented so far, working on this project provided me with valuable insight into the field of AI and especially Agentic AI. This Bachelor's Thesis work has strengthened my interest to further explore the field of AI systems.

References

- [1] Arena Intelligence 2026, ed. *Arena*. URL: <https://arena.ai/how-it-works> (visited on 07/29/2026) (cit. on p. 13).
- [2] Artificial Analysis. *LLM Leaderboard. Comparison of AI models from OpenAI, Anthropic, Google, SpaceXAI & others*. URL: <https://artificialanalysis.ai/leaderboards/models> (visited on 07/30/2026) (cit. on pp. 12, 15).
- [3] Artificial Analysis. *Artificial Analysis Long Context Reasoning Benchmark Leaderboard*. URL: <https://artificialanalysis.ai/evaluations/artificial-analysis-long-context-reasoning> (visited on 08/01/2026) (cit. on p. 15).
- [4] Artificial Analysis. *CritPt Benchmark Leaderboard*. URL: <https://artificialanalysis.ai/evaluations/critpt> (visited on 08/01/2026) (cit. on p. 15).
- [5] Artificial Analysis. *GPQA Diamond Benchmark Leaderboard*. URL: <https://artificialanalysis.ai/evaluations/gpqa-diamond> (visited on 08/01/2026) (cit. on p. 15).
- [6] Artificial Analysis, ed. *Independent analysis of AI*. URL: <https://artificialanalysis.ai/> (visited on 07/30/2026) (cit. on pp. 15, 21).
- [7] Artificial Analysis. *SciCode Benchmark Leaderboard*. URL: <https://artificialanalysis.ai/evaluations/scicode> (visited on 08/01/2026) (cit. on p. 15).
- [8] arXiv. *arXiv API Access*. 07/09/2026. URL: <https://info.arxiv.org/help/api/index.html> (cit. on pp. 16, 18).
- [9] Christian R. Ast, Jürgen Henk, Arthur Ernst, Luca Moreschini, Mihaela C. Falub, Daniela Pacilé, Patrick Bruno, Klaus Kern, and Marco Grioni. “Giant Spin Splitting through Surface Alloying”. In: *Phys. Rev. Lett.* 98 (18 05/2007), p. 186807. DOI: 10.1103/PhysRevLett.98.186807. URL: <https://link.aps.org/doi/10.1103/PhysRevLett.98.186807> (cit. on p. 28).
- [10] Florent Auvray, Jorge Puebla, Mingran Xu, Bivas Rana, Daisuke Hashizume, and Yoshichika Otani. “Spin accumulation at nonmagnetic interface induced by direct Rashba–Edelstein effect”. In: *Journal of Materials Science: Materials in Electronics* 29.18 (04/2018), pp. 15664–15670. ISSN: 1573-482X. DOI: 10.1007/s10854-018-9162-5. URL: <http://dx.doi.org/10.1007/s10854-018-9162-5> (cit. on p. 23).
- [11] Ajay Bandi, Bhavani Kongari, Roshini Naguru, Sahitya Pasnoor, and Sri Vidya Vilipala. “The Rise of Agentic AI: A Review of Definitions, Frameworks, Architectures, Applications, Evaluation Metrics, and Challenges”. In: *Future Internet* 17.9 (2025). ISSN: 1999-5903. DOI: 10.3390/fi17090404. URL: <https://www.mdpi.com/1999-5903/17/9/404> (cit. on pp. 6, 7).
- [12] M. Ben Shalom, A. Ron, A. Palevski, and Y. Dagan. “Shubnikov–De Haas Oscillations in SrTiO₃/LaAlO₃ Interface”. In: *Phys. Rev. Lett.* 105 (20 11/2010), p. 206401. DOI: 10.1103/PhysRevLett.105.206401. URL: <https://link.aps.org/doi/10.1103/PhysRevLett.105.206401> (cit. on p. 28).
- [13] Yu. A. Bychkov and E. I. Rashba. “Properties of a 2D electron gas with lifted spectral degeneracy”. In: *Письма в ЖЭТФ* 39 (2 1984), p. 66. URL: http://jetpletters.ru/ps/0/article_19121.shtml (cit. on p. 23).
- [14] Gaowei Chang. *Agent Network Protocol*. URL: <https://agent-network-protocol.com/> (visited on 07/29/2026) (cit. on p. 12).

- [15] Marina Chugunova, Dietmar Harhoff, Katharina Hölzle, Verena Kaschub, Sonal Malagimani, Ulrike Morgalla, and Robert Rose. “Who uses AI in research, and for what? Large-scale survey evidence from Germany”. In: *Research Policy* 55.2 (2026), p. 105381. ISSN: 0048-7333. DOI: <https://doi.org/10.1016/j.respol.2025.105381>. URL: <https://www.sciencedirect.com/science/article/pii/S0048733325002100> (cit. on p. 5).
- [16] CrewAI. *Agent/core*. URL: <https://github.com/crewAIInc/crewAI/blob/main/lib/crewai/src/crewai/agent/core.py> (visited on 07/30/2026) (cit. on p. 14).
- [17] CrewAI. *Agents*. URL: <https://docs.crewai.com/v1.15.11/en/concepts/agents> (visited on 08/05/2026) (cit. on pp. 17, 18).
- [18] CrewAI. *ArxivPaperTool*. URL: https://github.com/crewAIInc/crewAI/tree/main/lib/crewai-tools/src/crewai_tools/tools/arxiv_paper_tool (visited on 07/30/2026) (cit. on p. 16).
- [19] CrewAI. *crewai tools*. URL: https://github.com/crewAIInc/crewAI/tree/main/lib/crewai-tools/src/crewai_tools (visited on 07/30/2026) (cit. on p. 14).
- [20] CrewAI. *Crews*. URL: <https://docs.crewai.com/v1.15.16/en/concepts/crews> (visited on 07/29/2026) (cit. on p. 14).
- [21] CrewAI. *Flows*. URL: <https://docs.crewai.com/v1.15.9/en/concepts/flows> (visited on 07/30/2026) (cit. on pp. 14, 32).
- [22] CrewAI. *Tasks*. URL: <https://docs.crewai.com/v1.15.9/en/concepts/tasks#asynchronous-execution> (visited on 07/30/2026) (cit. on p. 14).
- [23] crewai. *crew.py*. 04/22/2026. URL: <https://github.com/crewAIInc/crewAI/blob/main/lib/crewai/src/crewai/crew.py> (visited on 04/22/2026) (cit. on pp. 14, 18, 32).
- [24] crewai. *crewAi*. 04/22/2026. URL: <https://crewai.com/> (visited on 04/22/2026) (cit. on pp. 13, 14).
- [25] crewai. *task.py*. 04/22/2026. URL: <https://github.com/crewAIInc/crewAI/blob/main/lib/crewai/src/crewai/task.py> (visited on 04/22/2026) (cit. on p. 13).
- [26] critPt. *challenges*. 04/22/2026. URL: https://github.com/CritPt-Benchmark/CritPt/tree/main/data/public_test_challenges (visited on 04/22/2026) (cit. on pp. 13, 21).
- [27] Linux Foundation AI & Data. *Agent Communication Protocol*. URL: <https://agentcommunicationprotocol.dev/about/mission-and-team> (visited on 07/29/2026) (cit. on p. 12).
- [28] Gesellschaft für wissenschaftliche Datenverarbeitung mbH Göttingen, ed. *SAIA*. URL: <https://docs.hpc.gwdg.de/services/ai-services/saia/index.html#api-request> (visited on 07/30/2026) (cit. on pp. 15–18, 30).
- [29] Hana Derouiche, Zaki Brahmi, and Haithem Mazeni. *Agentic AI Frameworks: Architectures, Protocols, and Design Challenges*. 2025. arXiv: 2508.10146 [cs.AI]. URL: <https://arxiv.org/abs/2508.10146> (visited on 07/28/2026) (cit. on pp. 12–14).
- [30] Tu Anh Dinh, Carlos Mullov, Leonard Bärman, et al. *SciEx: Benchmarking Large Language Models on Scientific Exams with Human Expert Grading and Automatic Grading*. 2024. arXiv: 2406.10421 [cs.CL]. URL: <https://arxiv.org/abs/2406.10421> (cit. on p. 13).
- [31] Ali Doosthosseini, Jonathan Decker, Hendrik Nolte, and Julian Kunkel. “SAIA: a seamless Slurm-native solution for HPC-based services”. In: *The Journal of Supercomputing* 82.7 (05/2026), p. 403. ISSN:

- 1573-0484. DOI: 10.1007/s11227-026-08508-3. URL: <https://doi.org/10.1007/s11227-026-08508-3> (cit. on pp. 15, 18).
- [32] Yogesh K. Dwivedi, Mohamed Y. I. Helal, Ibrahim A. Elgendy, Rasha Alahmad, Paul Walton, Ayoung Suh, Vinay Singh, and Il Jeon. “Agentic AI Systems: What It Is and Isn’t”. In: *Global Business and Organizational Excellence* 45.3 (2026), pp. 253–263. DOI: <https://doi.org/10.1002/joe.70018>. eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1002/joe.70018>. URL: <https://onlinelibrary.wiley.com/doi/abs/10.1002/joe.70018> (visited on 07/28/2026) (cit. on p. 12).
- [33] Victor M. Edelstein. “Spin polarization of conduction electrons induced by electric current in two-dimensional asymmetric electron systems”. In: *Solid State Communications* 73 (1990), pp. 233–235. URL: <https://api.semanticscholar.org/CorpusID:121468791> (cit. on p. 23).
- [34] Abul Ehtesham, Aditi Singh, Gaurav Kumar Gupta, and Saket Kumar. *A survey of agent interoperability protocols: Model Context Protocol (MCP), Agent Communication Protocol (ACP), Agent-to-Agent Protocol (A2A), and Agent Network Protocol (ANP)*. 2025. arXiv: 2505.02279 [cs.AI]. URL: <https://arxiv.org/abs/2505.02279> (visited on 07/29/2026) (cit. on p. 12).
- [35] Hans-Andreas Engel, Emmanuel I. Rashba, and Bertrand I. Halperin. “Out-of-Plane Spin Polarization from In-Plane Electric and Magnetic Fields”. In: *Physical Review Letters* 98.3 (01/2007). ISSN: 1079-7114. DOI: 10.1103/physrevlett.98.036602. URL: <http://dx.doi.org/10.1103/PhysRevLett.98.036602> (cit. on pp. 23, 24).
- [36] Motohiko Ezawa. “Out-of-plane Edelstein effects: Electric field induced magnetization in wave magnets”. In: *Physical Review B* 111.16 (04/2025). ISSN: 2469-9969. DOI: 10.1103/physrevb.111.1161301. URL: <http://dx.doi.org/10.1103/PhysRevB.111.L161301> (cit. on pp. 23, 24).
- [37] Takumi Funato and Mamoru Matsuo. “Acoustic Rashba–Edelstein effect”. In: *Journal of Magnetism and Magnetic Materials* 540 (12/2021), p. 168436. ISSN: 0304-8853. DOI: 10.1016/j.jmmm.2021.168436. URL: <http://dx.doi.org/10.1016/j.jmmm.2021.168436> (cit. on pp. 23, 24).
- [38] Irene Gaiardoni, Mattia Trama, Alfonso Maiellaro, Claudio Guarcello, Francesco Romeo, and Roberta Citro. “Edelstein Effect in Isotropic and Anisotropic Rashba Models”. In: *Condensed Matter* 10.1 (03/2025), p. 15. ISSN: 2410-3896. DOI: 10.3390/condmat10010015. URL: <http://dx.doi.org/10.3390/condmat10010015> (cit. on pp. 23, 24, 27).
- [39] Pietro Gambardella and Ioan Miron. “Current-induced spin-orbit torques”. In: *Philosophical transactions. Series A, Mathematical, physical, and engineering sciences* (08/13/2011). DOI: 10.1098/rsta.2010.0336 (cit. on p. 23).
- [40] Dan Hendrycks, Collin Burns, Steven Basart, Andy Zou, Mantas Mazeika, Dawn Song, and Jacob Steinhardt. *Measuring Massive Multitask Language Understanding*. 2021. arXiv: 2009.03300 [cs.CY]. URL: <https://arxiv.org/abs/2009.03300> (visited on 07/29/2026) (cit. on p. 12).
- [41] HuggingFace. *Agent Arena*. URL: <https://huggingface.co/spaces/lmarena-ai/arena-leaderboard> (visited on 07/28/2026) (cit. on p. 9).
- [42] HuggingFace. *Artificial Analysis LLM Leaderboard*. URL: <https://huggingface.co/spaces/ArtificialAnalysis/LLM-Performance-Leaderboard> (visited on 07/28/2026) (cit. on p. 9).
- [43] HuggingFace. *GLM-4.7-FP8*. URL: <https://huggingface.co/zai-org/GLM-4.7-FP8> (visited on 07/30/2026) (cit. on p. 15).
- [44] *Imprint*. URL: <https://gwdg.de/imprint/> (visited on 07/30/2026) (cit. on p. 15).

- [45] Artificial Intelligence. *CritPt Benchmark Evaluation API*. URL: <https://artificialanalysis.ai/api-reference#critpt-evaluate> (visited on 08/07/2026) (cit. on p. 21).
- [46] Jens Jacobi. “Ein Pascal-orientiertes Unterstützungssystem für grafische Darstellungen und Verarbeitung von Meßdaten in der Biotechnologie”. Diplomarbeit. Ingenieurhochschule Köthen
Sektion Biotechnologie, Lebensmitteltechnologie
Sektion Mathematik, Informatik, Rechentechnik, 01/1990 (cit. on p. 6).
- [47] Xiaotong Ji, Rasul Tutunov, Matthieu Zimmer, and Haitham Bou-Ammar. *Decoding as Optimisation on the Probability Simplex: From Top-K to Top-P (Nucleus) to Best-of-K Samplers*. 2026. arXiv: 2602.18292 [cs.LG]. URL: <https://arxiv.org/abs/2602.18292> (visited on 07/27/2026) (cit. on pp. 4, 7–9).
- [48] Jared Kaplan, Sam McCandlish, Tom Henighan, Tom B. Brown, Benjamin Chess, Rewon Child, Scott Gray, Alec Radford, Jeffrey Wu, and Dario Amodei. *Scaling Laws for Neural Language Models*. 2020. arXiv: 2001.08361 [cs.LG]. URL: <https://arxiv.org/abs/2001.08361> (visited on 07/28/2026) (cit. on pp. 9, 10).
- [49] Mojtaba Komeili, Kurt Shuster, and Jason Weston. *Internet-Augmented Dialogue Generation*. Ed. by Smaranda Muresan, Preslav Nakov, and Aline Villavicencio. Dublin, Ireland, 05/2022. DOI: 10.18653/v1/2022.acl-long.579. URL: <https://aclanthology.org/2022.acl-long.579/> (visited on 07/28/2026) (cit. on p. 10).
- [50] Vikram Krishnamurthy. *LLMs as High-Dimensional Nonlinear Autoregressive Models with Attention: Training, Alignment and Inference*. 2026. arXiv: 2602.00426 [cs.LG]. URL: <https://arxiv.org/abs/2602.00426> (cit. on p. 6).
- [51] Siddhant Kulkarni and Yukta Kulkarni. *Benchmarking Multi-Agent LLM Architectures for Financial Document Processing: A Comparative Study of Orchestration Patterns, Cost-Accuracy Tradeoffs and Production Scaling Strategies*. 2026. arXiv: 2603.22651 [cs.AI]. URL: <https://arxiv.org/abs/2603.22651> (visited on 07/28/2026) (cit. on pp. 11, 32).
- [52] Emmy Liu, Amanda Bertsch, Lintang Sutawika, et al. *Not-Just-Scaling Laws: Towards a Better Understanding of the Downstream Impact of Language Model Design Decisions*. 2026. arXiv: 2503.03862 [cs.CL]. URL: <https://arxiv.org/abs/2503.03862> (visited on 07/28/2026) (cit. on p. 10).
- [53] David Louapre, Joel Niklaus, and Lewis Tunstall. “physics-intern: an autonomous agentic framework for physics research”. In: *Hugging Face* (04/12/2026). URL: <https://huggingface.co/spaces/huggingface/physics-intern> (visited on 07/24/2026) (cit. on pp. 5, 32).
- [54] Chris Lu, Cong Lu, Robert Tjarko Lange, Jakob Foerster, Jeff Clune, and David Ha. *The AI Scientist: Towards Fully Automated Open-Ended Scientific Discovery*. 2024. arXiv: 2408.06292 [cs.AI]. URL: <https://arxiv.org/abs/2408.06292> (cit. on p. 5).
- [55] Rahul Marchand, Art O Cathain, Jerome Wynne, Philippos Maximos Giavridis, Sam Devere, John Wilkinson, Jason Gwartz, and Harry Coppock. *Quantifying Frontier LLM Capabilities for Container Sandbox Escape*. 2026. arXiv: 2603.02277 [cs.CR]. URL: <https://arxiv.org/abs/2603.02277> (visited on 07/30/2026) (cit. on pp. 10, 17).
- [56] Physics Modelling Assistant. Ed. by Janis Jacobi. URL: https://github.com/ceterumCenseoC/publicFiles/blob/main/version_1/information.md (cit. on p. 23).
- [57] Physics Modelling Assistant. Ed. by Janis Jacobi. URL: https://github.com/ceterumCenseoC/publicFiles/blob/main/version_1/model.md (cit. on p. 25).

- [58] Physics Modelling Assistant. Ed. by Janis Jacobi. URL: https://github.com/ceterumCenseoC/publicFiles/blob/main/version_1/unit_check.md (cit. on p. 26).
- [59] Physics Modelling Assistant. Ed. by Janis Jacobi. URL: https://github.com/ceterumCenseoC/publicFiles/blob/main/version_1/parameter_suggestion.md (cit. on p. 28).
- [60] Physics Modelling Assistant. Ed. by Janis Jacobi. URL: https://github.com/ceterumCenseoC/publicFiles/blob/main/version_1/simulation_physician.py (cit. on p. 29).
- [61] Physics Modelling Assistant. Ed. by Janis Jacobi. URL: https://github.com/ceterumCenseoC/publicFiles/blob/main/version_1/simulation_correction.py (cit. on p. 29).
- [62] Physics Modelling Assistant. Ed. by Janis Jacobi. URL: https://github.com/ceterumCenseoC/publicFiles/blob/main/version_1/final_output.md (cit. on p. 29).
- [63] Physics-Modelling-Assistant. Ed. by Janis Jacobi. URL: <https://github.com/ceterumCenseoC/publicFiles/blob/main/sources.md> (cit. on p. 23).
- [64] E. I. Rashba and V. I. Sheka. “Electric-Dipole Spin Resonances”. In: (2018). arXiv: 1812.01721 [cond-mat.mes-hall]. URL: <https://arxiv.org/abs/1812.01721> (cit. on p. 23).
- [65] David Rein, Betty Li Hou, Asa Cooper Stickland, Jackson Petty, Richard Yuanzhe Pang, Julien Dirani, Julian Michael, and Samuel R. Bowman. *GPQA: A Graduate-Level Google-Proof Q&A Benchmark*. 2023. arXiv: 2311.12022 [cs.AI]. URL: <https://arxiv.org/abs/2311.12022> (visited on 07/29/2026) (cit. on pp. 12, 19).
- [66] Stuart Russell and Peter Norvig. *Künstliche Intelligenz Ein moderner Ansatz*. Pearson Benelux B.V. (Zweigniederlassung Deutschland), 2023. ISBN: 9783868944303. URL: <https://elibrary.pearson.de/book/99.150005/9783863263263> (visited on 07/26/2026) (cit. on p. 6).
- [67] Erzhao Shao, Yifang Wang, Yifan Qian, Zhenyu Pan, Han Liu, and Dashun Wang. “SciSciGPT: advancing human–AI collaboration in the science of science”. In: *Nature Computational Science* (03/01/2026). URL: <https://doi.org/10.1038/s43588-025-00906-6> (visited on 07/24/2026) (cit. on pp. 5, 32).
- [68] Rao Surapaneni, Miku Jha, Michael Vakoc, and Todd Segal. *Announcing the Agent2Agent Protocol (A2A)*. 04/09/2025. URL: <https://developers.googleblog.com/en/a2a-a-new-era-of-agent-interopability/> (visited on 07/29/2026) (cit. on p. 12).
- [69] InternAgent Team, Bo Zhang, Shiyang Feng, et al. *InternAgent: When Agent Becomes the Scientist – Building Closed-Loop System from Hypothesis to Verification*. 2025. arXiv: 2505.16938 [cs.AI]. URL: <https://arxiv.org/abs/2505.16938> (cit. on p. 5).
- [70] LCC) The Linux Foundation Project (LF Projects). *Model Context Protocol*. URL: <https://github.com/modelcontextprotocol> (visited on 07/29/2026) (cit. on p. 12).
- [71] Romal Thoppilan, Daniel De Freitas, Jamie Hall, et al. *LaMDA: Language Models for Dialog Applications*. 2022. arXiv: 2201.08239 [cs.CL]. URL: <https://arxiv.org/abs/2201.08239> (visited on 07/28/2026) (cit. on p. 10).
- [72] Hanchen Wang, Tianfan Fu, Yuanqi Du, et al. “Scientific discovery in the age of artificial intelligence”. In: *Nature* (08/01/2023). URL: <https://doi.org/10.1038/s41586-023-06221-2> (visited on 07/24/2026) (cit. on p. 5).
- [73] Xiaoxuan Wang, Ziniu Hu, Pan Lu, Yanqiao Zhu, Jieyu Zhang, Satyen Subramaniam, Arjun R. Loomba, Shichang Zhang, Yizhou Sun, and Wei Wang. *SciBench: Evaluating College-Level Scientific Problem-*

- Solving Abilities of Large Language Models*. 2024. arXiv: 2307.10635 [cs.CL]. URL: <https://arxiv.org/abs/2307.10635> (visited on 07/29/2026) (cit. on p. 12).
- [74] Yiyi Wang. *Markdown Preview Enhanced*. URL: <https://github.com/shd101wyy/vscode-markdown-preview-enhanced> (visited on 07/17/2026) (cit. on p. 25).
- [75] Sandro Wefel. *Rechnernetze und verteilte Systeme WiSe 2025/2026*. URL: https://mos.uni-halle.de/Download/Aktuell/GB/MH_13108_aktuell.pdf (visited on 07/30/2026) (cit. on p. 16).
- [76] Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed Chi, Quoc Le, and Denny Zhou. *Chain-of-Thought Prompting Elicits Reasoning in Large Language Models*. 2023. arXiv: 2201.11903 [cs.CL]. URL: <https://arxiv.org/abs/2201.11903> (visited on 08/01/2026) (cit. on p. 10).
- [77] Yutaro Yamada, Robert Tjarko Lange, Cong Lu, Shengran Hu, Chris Lu, Jakob Foerster, Jeff Clune, and David Ha. *The AI Scientist-v2: Workshop-Level Automated Scientific Discovery via Agentic Tree Search*. 2025. arXiv: 2504.08066 [cs.AI]. URL: <https://arxiv.org/abs/2504.08066> (cit. on p. 5).
- [78] Xiang Yue, Yuansheng Ni, Kai Zhang, et al. *MMMU: A Massive Multi-discipline Multimodal Understanding and Reasoning Benchmark for Expert AGI*. 2024. arXiv: 2311.16502 [cs.CL]. URL: <https://arxiv.org/abs/2311.16502> (visited on 07/29/2026) (cit. on p. 12).
- [79] Wayne Xin Zhao, Kun Zhou, Junyi Li, et al. *A Survey of Large Language Models*. 2026. arXiv: 2303.18223 [cs.CL]. URL: <https://arxiv.org/abs/2303.18223> (cit. on p. 6).
- [80] Minhui Zhu, Minyang Tian, Xiaocheng Yang, et al. *Probing the Critical Point (CritPt) of AI Reasoning: a Frontier Physics Research Benchmark*. 2025. arXiv: 2509.26574 [cs.AI]. URL: <https://arxiv.org/abs/2509.26574> (cit. on pp. 13, 19–21).

6 Appendix

7 Declaration of authorship

I hereby declare that I have written this thesis independently and without unauthorized assistance.

All sources and references used have been properly cited.

This thesis has not been submitted in the same or substantially similar form for any other degree or academic qualification.

Halle (Saale), 24th August 2026

Janis Felix Jacobi

8 Acknowledgement

The authors gratefully acknowledge the computing time granted by the KISSKI project. The calculations for this research were conducted with computing resources under the project 9e28b773-a710-4a17-a242-523e92c812de.

I want to express my gratitude to Professor Schröter for the opportunity to write my Bachelors Thesis under his supervision and his valuable suggestions regarding the direction of my work. My sincere thanks goes to my second supervisor, Professor Lounis, for his valuable feedback for my Thesis, especially regarding his suggestions on designing the unit check. I greatly appreciate the feedback of Moritz Winterott regarding the design process as well the proofreading of this Thesis.

I want to thank my parents for supporting me throughout my studies and I want to express my appreciation to my girlfriend being with me through all frustrations, confusions, and delights experienced in the process.